

Functional connectivity, structural connectivity, and inter-individual variability in *Drosophila melanogaster*

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eLife Assessment

This paper presents a collection of analyses relating structure and function in the whole-brain *Drosophila* EM connectome and whole-brain calcium imaging data. The linkage of detailed anatomical structure with population activity is of broad interest in circuit neuroscience in light of increasingly detailed brain maps, but the analysis methods used made the evidence **incomplete**. The conclusions are **useful** for specific network observations, but a more thorough analysis of the anatomical and functional data is needed to support the overall claims.

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Abstract

Clarifying the relationship between structure and function is important for understanding the brain. In *Drosophila melanogaster*, whole-brain and hemibrain electron microscopy connectome data, as well as whole-brain calcium imaging data are available. We applied pre-processing methods from fMRI to whole-brain calcium imaging data and comprehensively investigated the optimal parameters. Then, we found that the FC-SC (functional and structural connectivity) correlation decreased linearly with ROI count, and this trend was the same in flies and humans. We also developed a new, more robust method to quantify the degree of pre- and post-synaptic segregation and investigated this in the fly whole-brain. This revealed that many neurons have unsegregated synapses. We extracted highly unsegregated synapses and compared them with random-extracted null SC matrices. Their FC-SC correlation was significantly higher, indicating that these synapses contribute to FC well. Conversely, highly segregated-synapses showed significantly lower FC-SC correlation and contribute less to FC. Therefore, neurons with unsegregated synapses like non-spiking neurons are spread throughout the whole brain, and they are thought to have a significant influence on FC.

Introduction

Functional connectivity (FC) describes the relationship between functional activity in one brain region and that in another. In previous approaches, two spike trains were analyzed by a cross-correlation histogram [1, 2]; then, this method was applied to positron emission tomography and functional magnetic resonance imaging (fMRI) [3]. Subsequently, the time-series correlation between two regions has become widely used as a measure of undirected FC [4]. Underlying such functional connectivity, a structural connectivity (SC) substrate exists, and its relationship with FC has been investigated using various modalities [5, 6, 7, 8, 9, 10, 11]. In humans, FC-SC correlations between resting-state (rs-) fMRI and diffusion tensor imaging have been explored, and studies have shown high relationships of $r=0.66$ for 66 regions and $r=0.54$ for 998 regions [6, 7]. Other studies have examined the relationship between rs-fMRI and SC based on retrograde tracing from 145 injection sites in the left cortex of marmosets ($r=0.379$ for 9862 voxels, **Ext.Data.Fig.2-1c**) [10] and the relationship between calcium imaging and SC based on electron microscopy (EM) in the hemibrain of *Drosophila melanogaster* (*D. melanogaster*) [12] ($r=0.74$ for 37 regions) [9]. Because of the disparity in species, number of ROIs, and modalities, a direct comparison across them becomes difficult. Therefore, in this study, we used whole-brain [13] and hemibrain [12] EM connectome data, which are currently considered to be the most detailed, and whole-brain calcium imaging data [14] of *D. melanogaster* to comprehensively investigate the relationship between FC and SC across various ROI shapes and numbers. Our results of the FC-SC correlations in the fly hemibrain compared to the human cortex at various ROIs showed roughly 1.08–1.24 times higher FC-SC correlations in the fly (**Fig. 2d**), and no large divergence was observed between the two species.

Because of the strong relationship between FC and SC, they can be used to investigate inter-individual variability in *D. melanogaster*. In particular, the FlyWire whole-brain and FlyEM hemibrain connectome data have been compared [15, 16], and widely different results are observed with cell typing [16]. However, these studies have yet to clarify whether these differences in connectome between the two individuals are attributed to inter-individual variability or some other reasons. Therefore, we compared these two EM connectomes based on the averaged FC of eight flies. FlyEM was scanned at a resolution of $8 \times 8 \times 8$ nm, and the connectome data has a “confidence” threshold that indicates the degree of synaptic detection. Conversely, FlyWire was scanned at a resolution of $4 \times 4 \times 40$ nm, and it had a “CleftScore” [17] threshold that shows the degree of synaptic detection. In fact, the thresholds initially employed by FlyWire and FlyEM caused large differences in post-synapse numbers; as such, we adjusted for the thresholds and compared the EM connectome data in each fly. We found that FlyEM clearly had higher values (more accuracy) than FlyWire in its FC-SC correlation and detection. In our judgment, this difference was not caused by inter-individual variability; instead, it was attributed to differences in the accuracy of detection and annotation of dendrites, axons, and synapses because of the voxel anisotropy.

We explored how functional activity contributed to the FC. In textbook terms, post-synapses on dendrites receive neurotransmitters and conduct an action potential along nerve fibers if they are excitatory; and pre-synapses on axons release neurotransmitters [18, 19]. In calcium imaging, the pre-synaptic activity in each region is detected as calcium signals, which represent functional activity. Then the FC is detected as the synchronization of functional activities. However, the synaptic arrangement on the neurons revealed in the EM connectome shows pre- and post-synapses that are highly intermingled in many neurons (i.e. input and output locations may be intermingled across axon and dendritic arbors); thus, signal transmission may be complex [19, 15, 13, 20]. Nerve cells include non-spiking neurons such as visual neurons, CT1 amacrine cells [21, 22, 23, 24, 25], interneurons in the antennal lobe [26], and APL cells in the mushroom body [27, 28, 29]. In *D. melanogaster*, these cells show localized

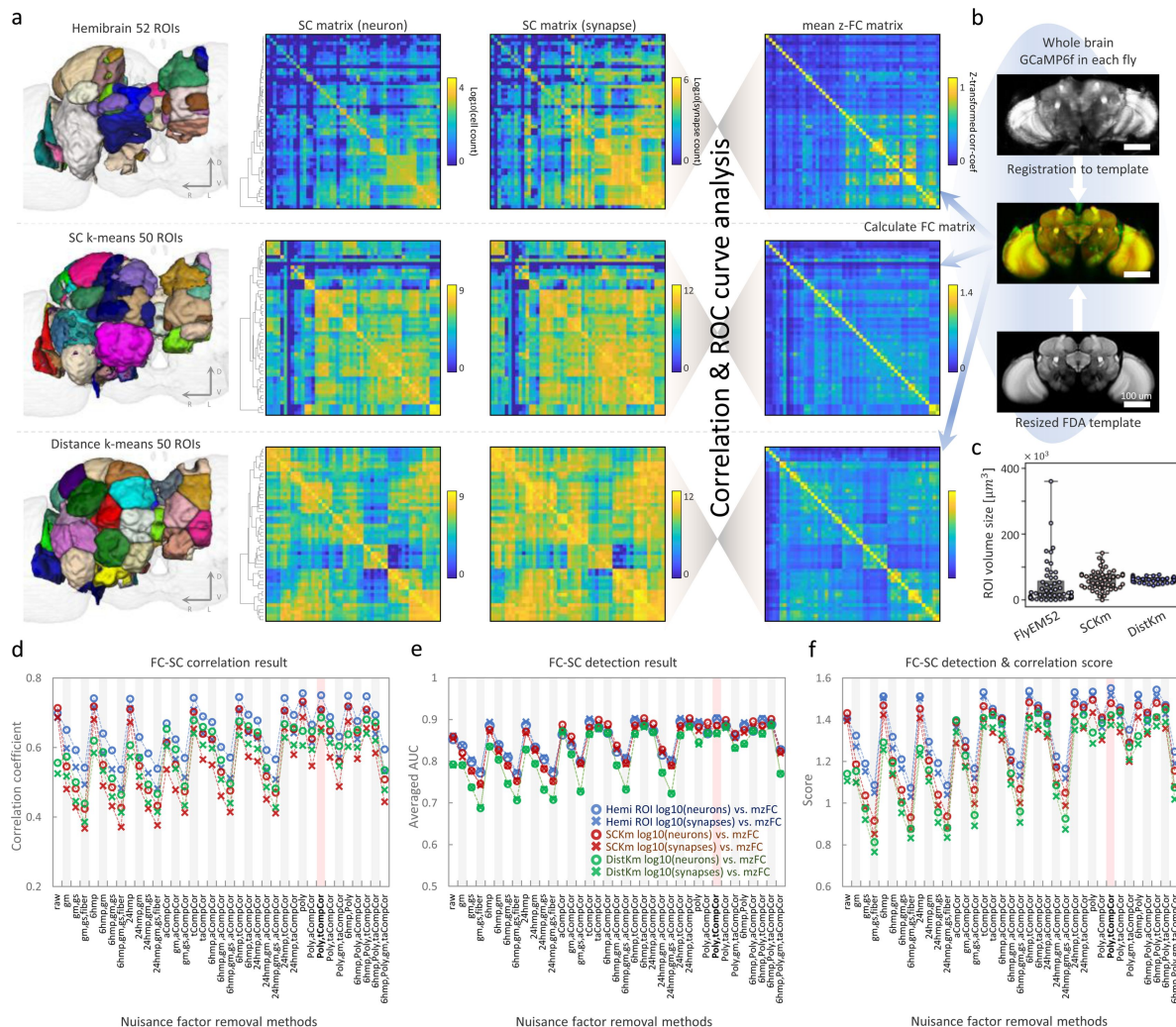


Fig. 1

Comparison between SC and FC, and optimal nuisance factor removal methods.

a, Schematic image of the comparison between SC matrices and FC matrices. (left) Three ROI types: Hemibrain anatomical 52 ROIs, SCKm 50 ROIs, and DistKm 50 ROIs. (center) SC matrices (neuron) and SC matrices (synapse) for each ROI type. (right) FC matrices for an average of 8 flies for each ROI type. **b**, Schematic image of calcium imaging data registration. **c**, Swarm plot of ROI volume for each ROI type. **d**, Results of nuisance factor removal method investigation. The vertical axis shows the correlation coefficient, and the horizontal axis shows the FC-SC correlation for each method. Results are plotted for the three ROI types. **e**, The vertical axis shows the averaged AUC of 100 thresholds, and the horizontal axis is the same as **d**. **f**, The vertical axis shows the FC-SC detection & correlation score, and the horizontal axis is the same as **d**.

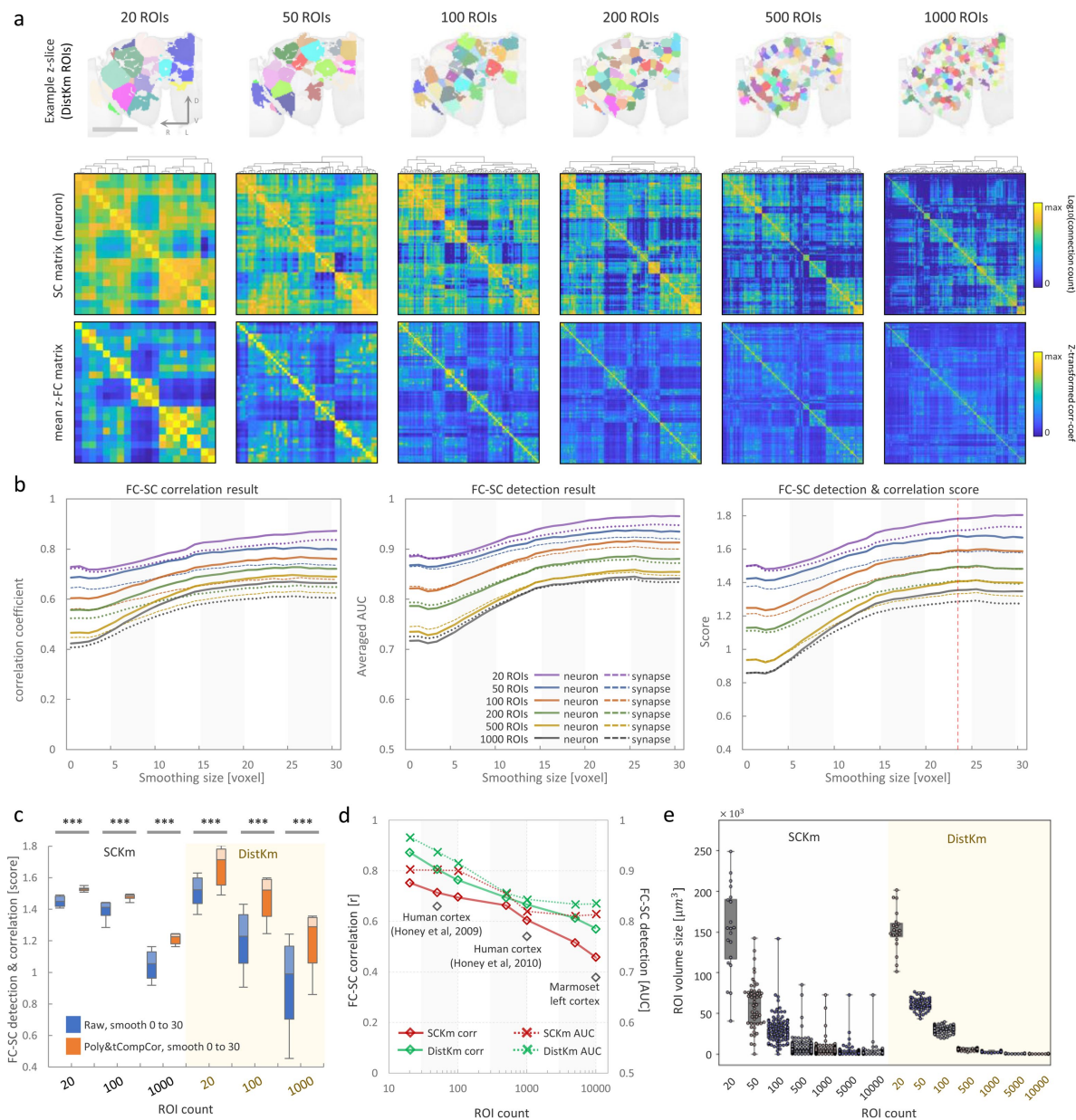


Fig. 2

Investigating spatial smoothing size, and comparison of FC-SC correlation between mammals and flies.

a, (top) Example z-slice of the reduced FDA template with DistKm ROIs from 20 to 1000 ROIs. (middle) SC matrices (neuron) of DistKm ROIs from 20 to 1000 ROIs. (bottom) FC matrices of DistKm ROIs from 20 to 1000 ROIs. **b**, From left to right, FC-SC correlation, FC-SC detection and FC-SC detection & correlation score, respectively. The solid line shows the FC vs. SC matrix (neuron), and the dashed line shows FC vs. SC matrix (synapse) at each ROI count. The vertical axis shows the correlation coefficient, averaged AUC and FC-SC detection & correlation score, respectively. The horizontal axis shows spatial smoothing size (voxels). **c**, Comparison between raw & smoothing (0 to 30 voxels) vs. Poly-tCompCor & smoothing (0 to 30 voxels) at each ROI count. Non-parametric Mann-Whitney U test was performed and Bonferroni correction was applied to correct for the familywise error rate (***) $p < 1.67 \times 10^{-4}$. **d**, Relation between FC-SC correlation and ROI count. (left vertical axis) Solid line shows FC-SC correlation (neuron) of SCKm and DistKm ROI type at each ROI count. (right vertical axis) The dashed line shows FC-SC detection (neuron) of SCKm and DistKm at each ROI count. **e**, Swarm plot of ROI volume of SCKm and DistKm at each ROI count.

functional activity taking place on a portion of the dendrite. In CT1 and APL cells, the pre- and post-synapses are highly intermingled (**Fig.4a** [↗](#), **Ext.Data.Fig.4-2** [↗](#)). Therefore, we wanted to investigate the presence of these unsegregated synapse-type neurons in the brain. We defined the pre- and post-synapse segregation index (PPSSI) and found that only some cells have a clear synaptic segregation, and many neurons have unsegregated synapses. When we extracted synapses with a particularly low segregation (high intermingling) and compared them between SC and FC, we found a significantly higher FC-SC correlation. Conversely, neurons with a high segregation had significantly lower FC-SC correlations. Many *Drosophila* brain neurons are interconnected [30 [↗](#)]. As such, we similarly investigated reciprocal synapses [31 [↗](#), 32 [↗](#), 33 [↗](#), 34 [↗](#), 35 [↗](#)], whose function is clearer than that of other unsegregated synapse types. When we extracted such reciprocal synapses, the FC-SC correlation was significantly higher. Our results indicated that unsegregated synapses contributed more to the FC. In other words, local activity due to unsegregated synapses could be considered the more essential functional activity. Unsegregated synapses are found not only in flies but also in mammals [31 [↗](#), 32 [↗](#), 18 [↗](#), 36 [↗](#), 37 [↗](#), 38 [↗](#)], including humans (**Ext.Data.Fig.1-2** [↗](#) from [39 [↗](#)]). Localized functional activity from pre-synaptic population dynamics can be observed by calcium imaging or detected through the neurovascular coupling with astrocytes [40 [↗](#)] via fMRI. Although modalities vary, the FC-SC correlation between humans and flies does not considerably differ, as shown earlier. Since various fMRI pre-processing and analysis methods can be directly applied to pre-synaptic population dynamics in this study, functional activity is locally compartmentalized in the whole brain, which is common among various species.

Results

Investigation of the optimal nuisance factor removal method for calcium imaging

First, the optimal pre-processing of the calcium imaging data of *D. melanogaster* was investigated. The registration method used in fMRI was applied to the calcium image (**Ext.Data.Fig.1-1** [↗](#)) to register to the FDA template [14 [↗](#)] (**Fig. 1b** [↗](#)). Then, various nuisance factor removal methods (details in Methods) were applied to the ROI time-series to generate the FC matrices of eight flies (**Fig. 1a** [↗](#)). Three types of ROIs were used in the analysis: ROIs based on the anatomical neuropil, commonly used in *Drosophila* (hemibrain ROI), ROIs based on k-means clustering of the SC matrix of 146,604 voxels in the hemibrain (SCKm), and ROIs based on distance-based clustering of voxels in the hemibrain (DistKm). SCKm can provide ROIs close to the shape of the anatomical neuropil across various ROI counts (**Ext.Data.Fig.2-1a** [↗](#)), while DistKm can provide ROIs that are completely unrelated to the anatomical neuropil. Next, a correlation analysis and ROC curve analysis were performed using the SC matrices based on FlyEM and the FC matrices with these ROI types (**Fig. 1a** [↗](#)). **Fig. 1d-f** [↗](#) shows the FC-SC correlation results, FC-SC detection results, and the normalized total score of FC-SC detection and correlation (details in Methods), respectively. After the comparison of the raw (leftmost) and removal methods for various ROI types, the combination of polynomial detrending [41 [↗](#)] & tCompCor [42 [↗](#)] was found to yield the optimal results. The optimal removal methods were found to differ between fMRI [10 [↗](#)] and calcium imaging data, and these results are further described in the Discussion section.

Comparison of the FC-SC correlation between humans and flies

Spatial smoothing is useful for absorbing inter-individual variability and conducting second-level group analysis [10 [↗](#), 43 [↗](#)]. Here, we explored the effect of spatial smoothing (of the calcium imaging data) on various ROI counts in DistKm (**Fig. 2a** [↗](#)) and SCKm (**Ext.Data.Fig.2-1a** [↗](#)) ROIs. In DistKm, smoothing was highly effective, and the optimal kernel radius of the Gaussian filter was around 23 voxels (52.44 μm) for various ROI counts (**Fig. 2b** [↗](#)). In SCKm, smoothing was ineffective for ROI count below 100, while the kernel radius above 16 voxels (36.48 μm) was

effective for ROI count above 100 (**Ext.Data.Fig.2-1b**). SCKm has a larger variation in volume for each ROI count than DistKm (**Fig. 1c**, **Fig. 2e**); as such, the effect of spatial smoothing is likely unstable. Next, we investigated whether the combination of spatial smoothing and Poly-tCompCor was effective. We found that this combination was significantly more effective than smoothing alone for SCKm and DistKm (**Fig. 2c**, Mann–Whitney U test $p < 1.67 \times 10^{-4}$ Bonferroni corrected). Conversely, the combination of a high-pass filter and nuisance factor removal overlapped for the purpose of noise removal, and the effect was degraded (**Ext.Data.Fig.2-1d**), as found in fMRI [10].

The optimal FC-SC correlation for various ROI counts was determined. Results showed that the correlation decreased proportionally to the logarithm of the ROI count (**Fig. 2d**). This was also compared between flies and mammals. At approximately 50 ROIs, the difference was nearly 1.08-fold to 1.22-fold ($r = 0.66$ for humans and $r = 0.71$ – 0.81 for flies [SCKm, DistKm]); by contrast, at approximately 1,000 ROIs, the difference was roughly 1.12-fold to 1.24-fold ($r = 0.54$ for humans and $r = 0.6$ – 0.67 for flies), which was not a considerable difference. The difference was slightly larger (1.21 to 1.5-fold: $r = 0.378$ for marmosets and $r = 0.46$ – 0.57 for flies) than the marmoset results at around 10,000 ROIs (**Ext.Data.Fig.2-1c**). Although SCs for flies and humans were obtained from single individuals, the tracer data is a composite from 52 marmosets [44], which might contribute to the difference with the fly's FC-SC correlation.

Comparison between FlyEM and FlyWire connectome data

We used the FlyEM's SC to compare the averaged FC of eight flies. In this section, we also used the FlyWire connectome data [45] to compare the FC and each SC, and possibly reveal the inter-individual variability of the two EM connectome data. As mentioned earlier, FlyEM has a synaptic “confidence” threshold and contains 21.4 million post-synapses in the hemibrain region at its initial value of 0. Conversely, FlyWire has a ‘CleftScore’ [17] threshold and 18.8 million post-synapses at its initial value of 50. Since the number of post-synapses differs at each threshold, we set the confidence threshold of 0.8 as high as possible, thereby increasing the FC-SC correlation (**Fig. 3d**, **ExtFig.3-1b**). Then we adjusted the CleftScore to 140, which has a post-synapse number of approximately 11.4 million (**Fig. 3a**). These thresholds were used for all analyses in this study. When we displayed the maximum Z-projection of the synapse point cloud of post-synapse counts in each voxel, we found that the trends in synapse detection completely varied (**Fig. 3b**). In each region, FlyEM had more synapses in the mushroom bodies (aL, bL, etc.) and in the major neuropils of the central nervous system (CNS; EB, FB, NO, and PB). In comparison, FlyWire had more synapses in the visual system (LO and ME) and in the periesophageal neuropils (CAN and FLA; **Fig. 3c**). In the FC-SC correlation of each region, FlyEM had higher values in most regions at 58:5 (**Fig. 3e**, **ExtFig.3-1a**). As a result, FlyEM $r = 0.73$ differed from FlyWire $r = 0.68$ at the matrix level (**Fig. 3d**). Both datasets were from genotype Canton-S-G1×w1118 adult females; FlyEM was five-day-old [12], and FlyWire was seven-day-old [13]. Although brain development may differ depending on the rearing environment, the tendency to detect synapses substantially varied. We attribute this to differences in detection accuracy resulting from the resolution of EM scanning, but not to inter-individual variability. We next used the FlyWire connectome data to investigate the ratios of neurotransmitter input (**ExtFig.3-2a**) and output (**Fig. 3f**) in each region. The highest percentage was acetylcholine ($62.8\% \pm 14.2\%$), followed by glutamine ($15.5\% \pm 9.2\%$), then GABA ($11.4\% \pm 8.2\%$) in neurotransmitter output. The analysis of the relationship between this ratio and FC-SC correlation in each neurotransmitter revealed significant correlations for acetylcholine ($r = 0.39$, $p = 0.0013$) and GABA ($r = -0.25$, $p = 0.046$). That is, the higher the percentage of excitatory connections, the higher the FC-SC correlation; conversely, the higher the percentage of inhibitory connections, the lower the FC-SC correlation. In humans, FC-SC correlations are lower in the evolutionarily expanded Brodmann area 39,45 and the temporal lobe [8]; therefore, these areas may have more long range inhibitory connections [46, 47, 48].

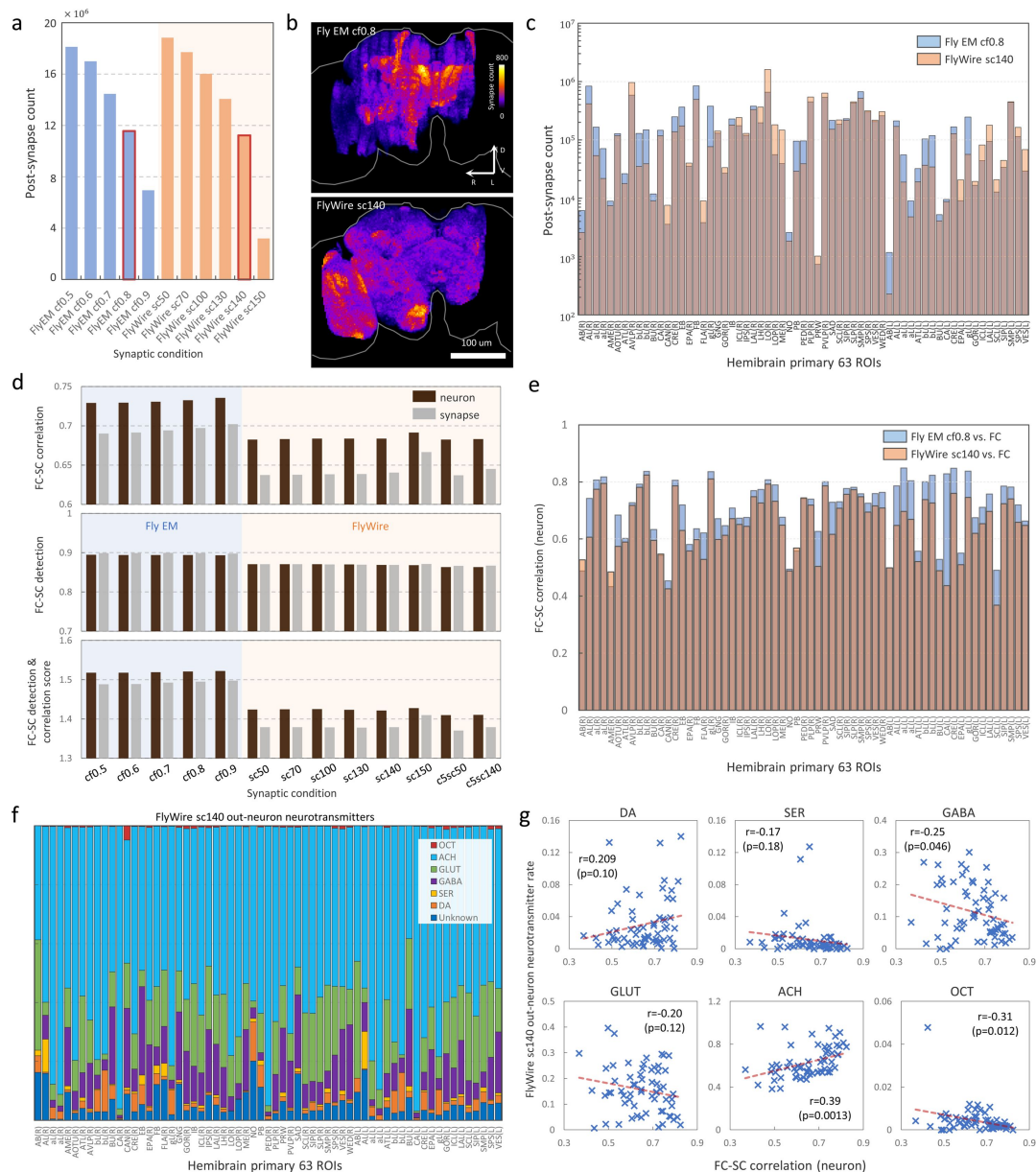


Fig. 3

Comparison between FlyEM and FlyWire connectome data.

a, Bar graph of post-synapse count in hemibrain region from FlyEM and FlyWire connectome data. The vertical axis shows post-synapse count, and the horizontal axis shows confidence thresholds for FlyEM and Cleft score thresholds for FlyWire. **b**, (top) Maximum Z-projection of synapse point cloud of FlyEM (confidence thresholds=0.8 [hereafter, cf0.8]) (bottom) Maximum Z-projection of synapse point cloud of FlyWire (Cleft score threshold=140 [hereafter, sc140]) **c**, Bar graph of post-synapse count in hemibrain primary ROIs. Blue bar shows FlyEM (cf0.8), red bar shows FlyWire (sc140). The vertical axis shows post-synapse count, and the horizontal axis shows hemibrain primary ROIs. **d**, Bar graphs of FC-SC correlation, FC-SC detection, and FC-SC detection & correlation score, respectively. The horizontal axis shows confidence thresholds for FlyEM and Cleft score thresholds for FlyWire. **e**, Bar graph of FC-SC correlation in hemibrain primary ROIs. Blue bars show FlyEM (cf0.8) vs. FC (neuron), red bars show FlyWire (sc140) vs. FC (neuron). The vertical axis shows FC-SC correlation, and the horizontal axis shows hemibrain primary ROIs. **f**, Bar graph of neurotransmitter rate of output neurons in hemibrain primary ROIs based on FlyWire (sc140) connectome data [45]. The horizontal axis shows hemibrain primary ROIs. (DA: dopamine, SER: serotonin, GABA, GLUT: glutamine, ACH: acetylcholine, OCT: octopamine) **g**, Scatter plots of neurotransmitter rate of output neurons vs. FC-SC correlation (neuron) in hemibrain primary ROIs (FlyWire sc140). Each of the six neurotransmitters was compared.

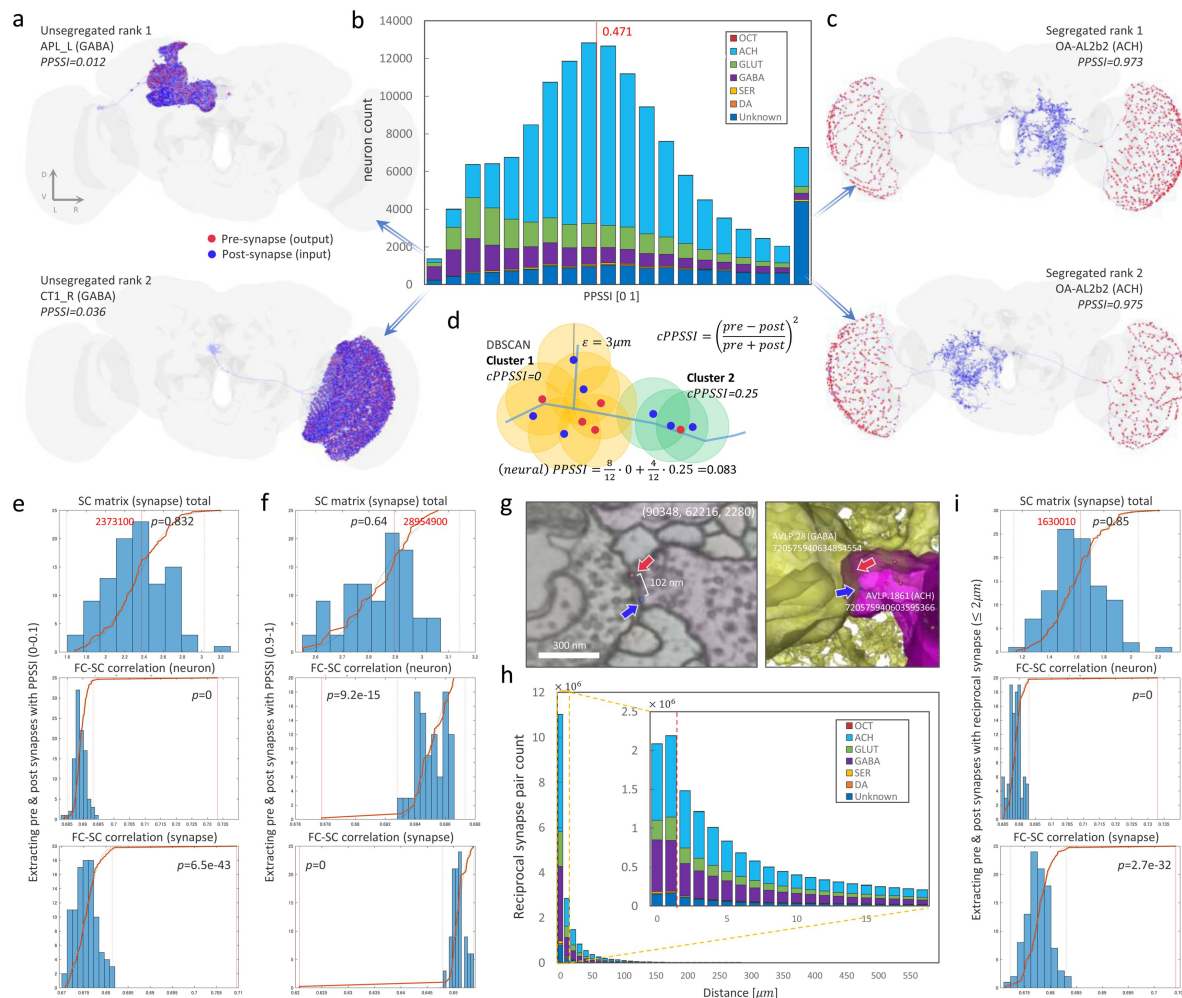


Fig. 4

Quantification of pre- and post-synapse segregation, and the relationship between FC and synapses.

a, Unsegregated first (APL_L cell) and second (CT1_R cell) ranking in FlyWire (sc140) connectome data. A red dot shows pre-synapse, a blue dot shows post-synapse. **b**, The PPSSI histogram of all neurons (139,255) in FlyWire (sc140) connectome data. The vertical axis shows neuron count, and the horizontal axis shows PPSSI [0 1]. **c**, Segregated first (OA-AL2b2 cell) and second (OA-AL2b2 cell) ranking in FlyWire (sc140) connectome data. **d**, Schematic image of pre- and post-synapse clustering based on DBSCAN, and calculation of cPPSSI and PPSSI. **e**, Histogram of null SC matrices (blue bar) and extracted SC matrix with PPSSI (0-0.1) (red solid line). Black dotted line shows cumulative distribution function of the normal distribution, brown solid line shows cumulative distribution function of null & extracted SC matrices, and red dotted line shows Bonferroni-corrected $p<0.05$ threshold. Top shows SC matrix (synapse) total, middle shows FC-SC correlation (neuron), and bottom shows FC-SC correlation (synapse). **f**, Histogram of null SC matrices and extracted SC matrix with PPSSI (0.9-1). **g**, Example of a reciprocal-synapse in the EM image (left) and in 3D (right). **h**, Histogram of reciprocal-synaptic minimum distances in FlyWire (sc140) connectome data. The vertical axis shows reciprocal-synapse pair count, and the horizontal axis shows distance [μm]. **i**, Histogram of null SC matrices and extracted SC matrix for reciprocal-synapses ($\leq 2\mu m$).

Unsegregated synapses contribute considerably to functional connectivity

We compared FC-SC in the macroscopic regions, and we also investigated the relationship between the synapses, the source of functional activity (pre-synaptic population dynamics), and FC. First, we examined the synaptic segregation in whole-brain neurons because many neurons showed pre- and post-synaptic intermingling. Then, we determined how unsegregated and segregated synapses affected the FC. To quantify synaptic segregation, we developed the PPSSI (**Fig. 4d**). Using DBSCAN [49] with a radius of 3 μm (details on Methods), we clustered synapses in each neuron and calculated the degree of segregation for each cluster (cPPSSI). We then multiplied the cPPSSI by the fraction of the synapse number and obtained the sum to determine the PPSSI of each neuron. **Fig. 4b** shows the result of applying PPSSI to 139,255 whole-brain neurons of FlyWire. The result showed a histogram resembling a normal distribution. When we created unsegregated ranking (details in Methods) by considering the number of synapses, APL and CT1 cells occupied the top 1–4 positions, indicating that we efficiently extracted unsegregated neurons (**Fig. 4a**, **Ext.Data.Fig.4-3a**). APL was also at the top for FlyEM (**Ext.Data.Fig.4-1a**). Similarly, we established segregated ranking and successfully extracted neurons with a high degree of segregation (**Fig. 4c**, **Ext.Data.Fig.4-3b**; **Ext.Data.Fig.4-1c** and **Ext.Data.Fig.4-4** for FlyEM).

Next, synapses with a cPPSSI between 0 and 0.1 were extracted, and the SC matrices (unsegregated synapses) of neurons and synapses were created. Synapses were then randomly sampled to generate 99 null SC matrices (random synapses), centered around the total number of connections (2,373,100) in the SC matrix (unsegregated synapses), forming a normal distribution (**Fig. 4e**, $p = 0.832$, details in Methods). The FC-SC correlations of the SC matrices (unsegregated synapses) of neurons and synapses were significantly higher than those of the randomly extracted samples (**Fig. 4e**, $p = 0$, $6.5\text{e-}43$; **Ext.Data.Fig.4-1d** for FlyEM). Therefore, unsegregated synapses might strongly contribute to FC. A similar analysis was performed on synapses with a cPPSSI of 0.9–1, which surprisingly showed a significantly lower FC-SC correlation (**Fig. 4f**, $p = 9.2\text{e-}14$, 0; **Ext.Data.Fig.4-1e** for FlyEM). Therefore, segregated synapses may not be able to contribute to strongly to the FC, and these findings are further explored in the Discussion section.

Among unsegregated synapses, reciprocal synapses have been well studied [31, 32, 33, 34, 35] and their function is considered to be clear (**Fig. 4g**, **Ext.Data.Fig.4-1f**). The shortest distance of such reciprocal synapses was extracted from whole-brain neurons, shown in **Fig. 4h** (**Ext.Data.Fig.4-1g** for FlyEM). They are particularly abundant within 2 μm (**Ext.Data.Fig.4-2a,c** for CT1 and APL cells), and acetylcholine and GABA constitute the majority (**Fig. 4h**). The comparison of the SC matrix based on reciprocal synapses ($<2 \mu\text{m}$) with the null SC matrices showed that the FC-SC correlation was significantly higher (**Fig. 4i**, $p = 0$, $2.7\text{e-}32$; **Ext.Data.Fig.4-1h** for FlyEM). Therefore, unsegregated synapses and reciprocal synapses ($<2 \mu\text{m}$) strongly contributed to FC. Furthermore, since the FC was the average of eight flies, such synapses are likely functionally common among individuals.

Region suppression analysis during fly walking

The whole-brain calcium imaging dataset from Brezovec et al. [14] provides fly walking data. Since fMRI pre-processing methods can be applied to calcium imaging data, we investigated whether voxel-level analysis using a general linear model (GLM) [50] could be applied to the fly walking task (**Fig. 5a**, details in Methods). **Fig. 5b** shows the GLM analysis result. Similar to the results of Brezovec et al., our findings revealed that LAL, FB, and IPS neuropils showed strong activity; the GNG neuropil, which is highly connected to leg neuropils in the ventral nerve cord [51], also exhibited strong activity. Although Brezovec et al. detected voxels that are strongly correlated with velocity, the GLM extracted voxels associated with walking behavior. Conversely,

we found inactivity or strong suppression in the Wedge and AVL P neuropils. We searched for a GABA neuron that would fit with this phenomenon and managed to find one that extended its dendrites widely from the Wedge to AVL P neuropils (**Fig. 5e, h**). The synapse point cloud could also fit in this region (**Fig. 5c**). We called this neuron the Wedge-AVL P GABA neuron (WAGN). The Wedge neuropil receives mechanosensory (wind, sound) input and connects to WPNb neurons (**Ext.Data.Fig.5-1a,b**), which then connect to ER1 ring neurons that act as the fly's compass [52, 53]. Interestingly, the left WAGN was strongly interconnected with WPNb tier 2/3 [54] and received input from 22 ascending neurons (ANs; **Fig. 5d**). Therefore, it may be a pathway that provides walking behavior feedback to the compass; however, wind caused by walking was absent in the experimental setup (the fly was fixed and walked on a treadmill), and may relate to the strong regional suppression that was observed. As such, further experimentation is needed to pursue this topic.

The left WAGN resembles a hub neuron; however, like CT1 and APL cells, it has highly unsegregated synapses (PPSSI=0.133). Synapses on the medial side were highly segregated, but the lateral side was well unsegregated (**Fig. 5h**). It received input from ANs (212 synapses) mainly on the ventral side, and outputs to WPNb (65 synapses), widely from the medial to the lateral side (**Fig. 5e**). A histogram of the shortest straight-line distances between pre-synapses to WPNb and post-synapses from ANs is shown in **Fig. 5f**, and the closest distance was 852.3 nm (**Fig. 5g**). Many acetylcholine inputs (disynaptic inhibition [25]) were closer than this value (**Fig. 5i**), and many GABA inputs from ANs (**Fig. 5f**) further suppress such disynaptic inhibition. AVL P ventral suppression (**Fig. 5b** white arrows) may be the suppression from these ANs. Thus, a highly delicate and localized control is possibly exercised from WAGN to WPNb tier 2/3.

Discussion

Although nuisance factor removal methods such as global mean (GM) and aCompCor [42] are optimal for fMRI [10], calcium imaging showed different results. GM is the average of all voxels, and the regression of this signal is useful in fMRI because noise from the equipment and coils affects the entire voxel image. Conversely, in two-photon microscopy, noise does not affect all voxels because of the excitation of specific molecules by laser absorption, and GM does not show a positive effect. aCompCor extracts the component time-series from the voxel time-series of white matter and cerebrospinal fluid (CSF) via PCA and then performs a regression. In flies, the CSF had no ROI mask, but fibers had ROIs; therefore, fibers were used instead of the white matter. However, many synapses may exist around these fibers, and aCompCor was not as effective as in humans. Conversely, polynomial detrending [41] was highly effective (**Fig. 1d-f**), likely because of linear thermal noise and other factors caused by the continuous excitation of molecules (for 30 min). tCompCor [42] is the component time-series extracted by PCA from voxels with the top 1000 standard deviations among all voxels. It potentially captures minute fluctuations caused by fly movements and effectively removes them.

The SC matrix from a cPPSSI of 0.9–1 synapses resulted in a significantly lower FC-SC correlation than the SC matrix from randomly selected synapses (**Fig. 4f** for FlyWire, **Ext.Data.Fig.4-1e** for FlyEM). This result was surprising, but we propose several possible reasons. Neurons with highly segregated synapses include ascending neurons (ANs) and descending neurons (DNs), (**Ext.Data.Fig.4-3b**). They also have a small number of unsegregated synapses (i.e. Sep rank9: PPSSI=0.959, Sep rank11: PPSSI=0.964). They are thought to be specialized for input or output in the CNS regions and may show large differences between SC and FC. Another reason is that these neurons may have widespread logical OR inputs and outputs. In **Fig. 4c**, an action potential at one point in the input synapse (blue) is likely transmitted to all red synapses. Such an OR circuit-like disproportionate input–output possibly causes a mismatch with the FC, because the functional activity in the output region originates from activity at a single location. These neurons are likely

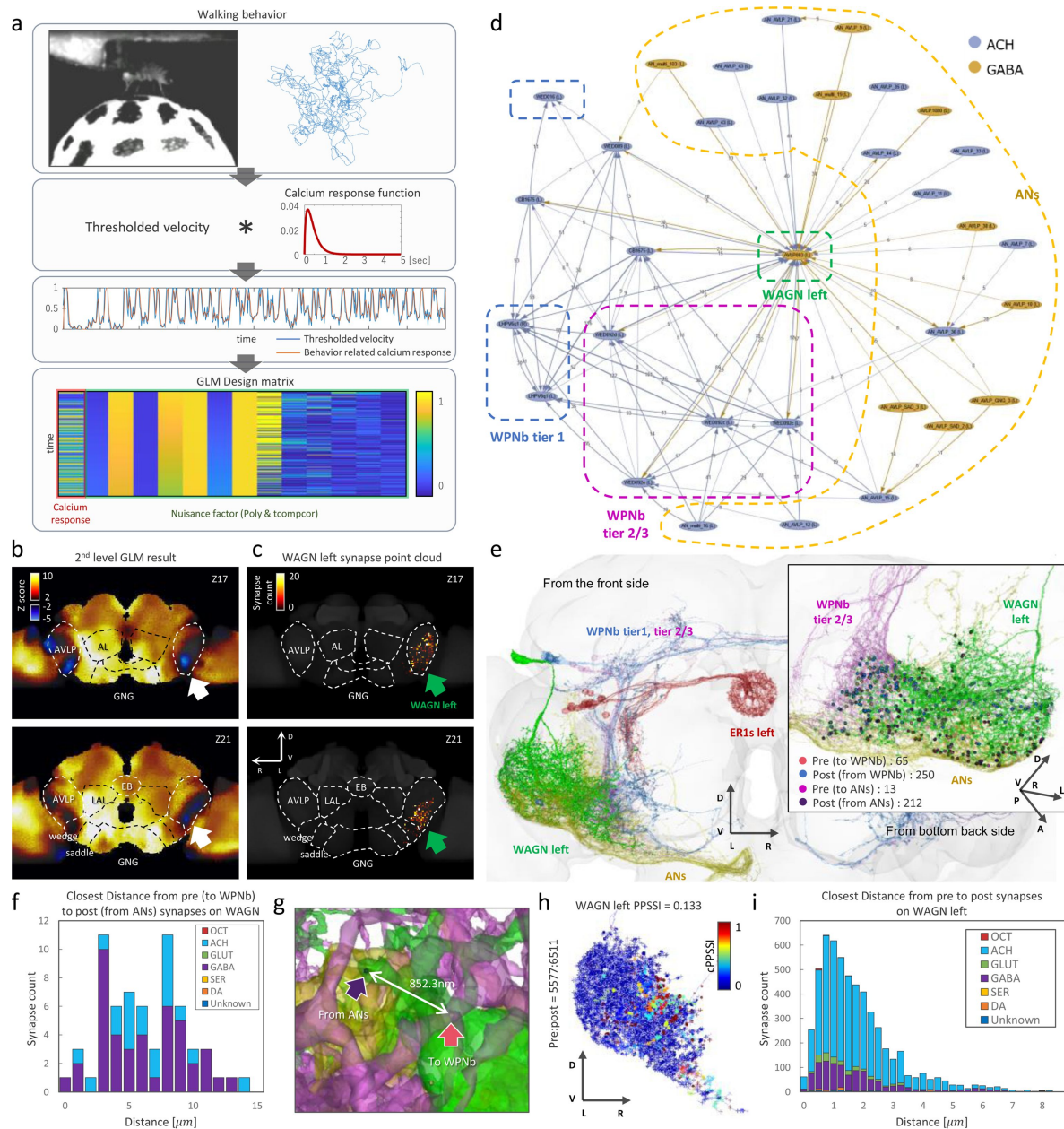


Fig. 5

General Linear Model analysis result and Wedge-AVLP GABA neuron.

a, Schematic image of GLM analysis. From the top, experimental image and walking trajectory, calcium response function, thresholded velocity, and GLM design matrix, respectively. **b**, 2nd level (group) GLM analysis result from 9 flies. Top shows Z=17 slice of the resized FDA template, and the bottom shows the Z=21 slice. The white arrow indicates the suppressed region during the fly walking task. **c**, Synapse point cloud of the left WAGN in FlyWire (sc140) connectome data. Top shows Z=17 slice of the resized FDA template, and bottom shows the Z=21 slice. **d**, Network graph of the left WAGN, ANs, WPNb tier 1 and tier 2/3 neurons from FlyWire codex (sc50, connection threshold>5). Blue is acetylcholine, ochre is GABA neuron. The small number on the edges shows synapse count. **e**, 3D image of the left WAGN, ANs, WPNb tier 1, tier 2/3 and left ER1 neurons, frontal view. Inside the black square shows the left WAGN, ANs and WPNb tier 2/3 from the back side. **f**, Histogram of closest distance from pre-synapse (to WPNb) to post-synapse (from ANs) in the left WAGN. Horizontal axis shows distance [μm]. Neurotransmitter indicates input ANs to post-synapses. **g**, Example 3D image of pre-synapse (to WPNb) to post-synapse (from ANs). **h**, All pre- and post-synapses of the left WAGN from the front. Color indicates cPPSSI in each synapse. **i**, Histogram of closest distance from pre-synapse to post-synapse in the left WAGN. The horizontal axis shows distance [μm]. Neurotransmitter indicates input neurons to post-synapses.

those whose FC are difficult to capture. Conversely, neurons with highly unsegregated synapses may capture the inputs of neighboring synapses, such as forming a logical AND circuit, correlating well with the output regions.

The FC-SC correlation was highly dependent on ROI count in both flies and humans (**Fig. 2d**). Through optimal nuisance factor removal and smoothing, we achieved $r=0.87$ at 20 ROIs. Furthermore, by applying Spearman correlation instead of Pearson, we obtained $r=0.928$. Thus, for a reasonably small ROI count the correlation was very high; however, for 10,000 ROIs, the correlation decreased to $r=0.57$. This phenomenon could be attributed to the presence of highly segregated synapses. Such neurons are widespread in the CNS (**Fig. 4b**, **Ext.Data.Fig.4-2f**) and are thought to provide logical OR circuits. When ROI volume was large (when the ROI count was small), a single ROI could cover the distributed input synapses. However, when a ROI was divided, input synapses were also divided across many ROIs, and the mismatch between SC and FC would increase. Other possibilities include the presence of gap junctions and FC due to neuropeptides [55], but they are outside of the scope of the present study and their relationship with ROI count is unclear.

In this study, we applied fMRI techniques directly to calcium imaging in flies and found many similarities between humans and flies (**Fig. 2d**). The calcium signal represents pre-synaptic population dynamics and likely includes the localized functional activity of unsegregated synapses [25, 31, 32, 33, 34], such as non-spiking [21, 22, 23, 24, 25, 26] and sub-threshold neuronal activity [18]. For fMRI, the signal captures localized functional activity as blood-oxygenation-level-dependent signal via neuro vascular coupling [40]. These two signals can be pre-processed (**Fig. 1**, **Fig. 2**) and analyzed (**Fig. 5a**) in surprisingly similar ways. This is unlikely a coincidence, and it suggests that similar design concepts underlie neural circuits in flies and mammals. Using *D. melanogaster* as a model animal, we were able to elucidate detailed local functional activities around unsegregated synapses and compare these results to the brain activities of mammals, including humans.

Materials and methods

Preprocessing of *D. melanogaster* calcium imaging data

Whole-brain calcium imaging data of *D. melanogaster* were acquired by Brezovec et al. [14] and downloaded from DANDI (<https://dandiarchive.org/dandiset/000727/0.240106.0043>). 4D NIfTI image data (256×128×49 voxels, 3384 frames) were extracted from a NWB file by an in-house MATLAB script. The FDA template, made from in-vivo calcium imaging, was resized (256×128×49 voxels, 2.45×2.28×3.72μm for a voxel) and used as a template because the original thresholded FDA template [56] was extremely large (1652×768×479 voxels), and the calcium imaging file was large (more than 40 GB); therefore, a smaller template with the imaging data was preferably used. Preprocessing and image registration were performed using Statistical Parametric Mapping (SPM12) [57] and ANTs [58] (**Ext.Data.Fig.1-1a**). Motion and slice timing correction were performed using SPM12, and the averaged NIfTI image was registered by ANTs to the resized FDA template. Lastly, 4D NIfTI images were transformed using the transform information. The data contained $n=9$ flies, but 8 flies were used for the ROI-based analysis because of missing time-series in some ROIs in one fly. Nine flies were used in the GLM analysis.

Combinations of the following nuisance factor removal methods were investigated: global mean (average signal across all voxels), global signal (average signal across all brain voxels), mean white matter (average signal across fiber voxels), 6HMP (head motion parameters), 24HMP, aCompCor, tCompCor [42], and polynomial detrending (Poly) [41]. These methods were implemented by an in-house MATLAB script [59]. A full width at half maximum (FWHM) from 1 voxel (2.28 μm)

to 30 voxels (68.4 μm), at a 1 voxel step, was investigated to determine the optimal spatial smoothing size for calcium imaging data. Temporal high-pass filters were also examined from 0.1 Hz to 0.001 Hz.

Transformation of EM connectome data into the resized FDA template

In the FlyWire case, connectome data, such as neurons and synapses, were downloaded from FlyWire Codex (<https://codex.flywire.ai/>) and Zenodo (<https://zenodo.org/records/10676866>) [13, 60, 45]. Detailed synapse data were saved in an Apache feather format, which was then converted to a CSV file. Synapse points were transformed from the FlyWire space to a JRC2018 female template [61] by the Python navis-flybrains package [61]. Since FlyWire coordinates are left and right reversed [13], they were flipped on the JRC2018 female template, and these points were transformed to the resized FDA template by ANTs. The use of these two templates facilitated precise data alignment across different modalities.

In the FlyEM case, the latest connectome data were downloaded from Janelia's DVID website (<https://dvid.io/blog/release-v1.2/>). Like those in the FlyWire case, synapse points were transformed from the FlyEM space to a JRC2018 female template by the Python navis-flybrains package. These points were transformed to a resized FDA template by ANTs.

Generation of regions of interest for the hemibrain

The hemibrain neuropil atlas (114 ROIs, including primary and sub-regions, aligned with the FlyEM space) was downloaded from Virtual Fly Brain (<https://www.virtualflybrain.org/>). ROI files were transformed from the FlyEM space to a JRC2018 female template by Java template-building scripts (<https://github.com/saalfeldlab/template-building>) [61]. They were transformed to a resized FDA template by ANTs. In Fig. 3, 63 primary ROIs mentioned above were used. The 52 ROIs in Fig. 1 were selected from the primary ROIs, and approximately 50 ROIs (those used by Turner et al. [9]) were selected. ROIs were segmented by SCKm (Fig. 1 and Fig. 2) via the k-means clustering of the SC matrix of 146,604 voxels in the hemibrain. They were also segmented by DistKm (Fig. 1 and Fig. 2) via the k-means clustering of the distance of 146,604 voxels in the hemibrain.

The SC matrix was calculated according to synapse locations and ROIs. Since synapse data formats differ between FlyEM and FlyWire, they were standardized by using post-synapses within ROIs. The input and output neurons at each ROI were extracted from the paired pre- and post-synapse data. Then the input and output neurons and their synapses that were common between ROI pairs were counted to create SC matrices for neurons and synapses. The FC matrix was calculated using the pre-processed voxel time-series and ROIs. The intensity of the voxels in the ROI was averaged to obtain the average intensity time-series of the ROI. Then, Pearson correlation was calculated between ROI pairs to generate the FC matrix. The correlation coefficients were z-transformed, calculated for eight flies, and then averaged to obtain the final FC matrix.

FC-SC detection and correlation

FC-SC correlation was calculated by taking the Pearson correlation between the logarithm of the SC matrix and the FC matrix, as done in other studies [6, 7, 10]. In the SC matrix, elements with zero connections become $-\text{Inf}$ when logarithmically converted, so we substituted zero for them. Also, the diagonal elements of the FC matrix become Inf when z-transformed, so we replaced them with the maximum value of the matrix. For one FC matrix, we calculated its correlation with both the neuron SC matrix and the synapse SC matrix. FC-SC detection was calculated by ROC (Receiver Operating Characteristic) curve analysis. This gives the detection accuracy of the FC matrix when the SC matrix is used as ground truth. Thresholds were taken from 0% to 99% by 1% steps of all elements of the SC matrix, and 100 AUCs (Area Under the Curve) were calculated. The

average of these AUCs is called the FC-SC detection. FC-SC correlation indicates the similarity of the two matrices, while FC-SC detection indicates the detection accuracy. Although the digit number of connections differs between neurons and synapses, FC-SC detection can absorb such differences. The FC-SC detection & correlation score was calculated as follows:

$$FC-SC_{det\&corr\ score} = |FC-SC_{correlation}| + 2 \times |FC-SC_{detection} - 0.5| \#(1)$$

The FC-SC detection & correlation score is in the range of [0 2], where the larger the score, the higher the similarity and detection accuracy.

PPSSI

The PPSSI (**Fig. 4d**) was used in DBSCAN [49] with a radius of 3 μm to generate pre- and post-synaptic clusters. The radius was empirically defined by several criteria. Firstly, CT1 and APL cells have the shortest distance of reciprocal synapses, within 2 μm (**Ext.Data.Fig.4-2a,c**). In zebrafish, reciprocal synapses were defined within 2.5 μm [35]. Schneider-Mizell et al. [19] defined a twig as a branch within 3 μm from the backbone of the cytoskeleton, and 80% of synaptic inputs in motor neurons are in the twig. In addition, 97% of pre-synaptic sites are located at less than 3 μm from the mitochondria. Therefore, 3 μm was deemed appropriate for the synapse cluster radius. Once the synapse clusters were determined by DBSCAN, the degree of synaptic segregation for each cluster was calculated using the following formula:

$$cPPSSI_i = \left(\frac{pre_i - post_i}{pre_i + post_i} \right)^2, \#(2)$$

where pre_i and $post_i$ are the pre- and post-synapse counts in cluster i , respectively. The $cPPSSI$ of each cluster (each synapse) is in the range of [0 1]. The PPSSI of each neuron was calculated as follows:

$$PPSSI = \sum_{i=1}^N \frac{(pre_i + post_i)}{W} \cdot cPPSSI_i \#(3)$$

where N is the cluster number, and W is the total synapse count for a neuron.

The unsegregated ranking was calculated based on the PPSSI and total synapse count of neuron k :

$$unsegregated\ score_k = \log_{10}(W_k) \cdot (1 - PPSSI_k) \#(4)$$

where $PPSSI_k$ is the PPSSI of neuron k and W_k is the total synapse count of neuron k . This allows us to extract neurons with many inputs and outputs and highly unsegregated synapses. The segregated ranking was also calculated in the same way, using the following score:

$$segregated\ score_k = \log_{10}(W_k) \cdot PPSSI_k \#(5)$$

A similar index to our PPSSI, namely, the segregation index, has been defined in the literature [19], but its use poses several problems. For neurons such as ANs and DNs, that have only input or output synapses, the segregation index results in calculations of $0 \times \infty$ or division by 0; consequently, calculation becomes impossible. Despite the [0 1] range of indices, the histogram is heavily biased toward 0 because of the use of the logarithm, and the mean is much closer to 0 than 0.5 (**Ext.Data.Fig.4-1i, j**). Therefore, in the present study, we developed the more robust PPSSI because it is less biased and its mean is close to 0.5 (**Fig. 4b**, **Ext.Data.Fig.4-1b**).

GLM Analysis for the fly walking task

GLM analysis (Fig. 5a) was performed to investigate the whole-brain functional activity during the fly walking task. Movement velocity data were extracted from an NWB file by an in-house MATLAB script. Walking was defined as 1σ (34%) or more of the movement velocity and divided into walking and other. Less than 10% of movement velocity was zero cut. The canonical calcium response function (CRF) was created by the gamma probability density function ($a=1.7$, $b=0.25$) in MATLAB based on the GCaMP6f response [62]. The walking-task time-series was transformed into a calcium response time-series by convolution with the CRF. GLM design matrices were created using the calcium response time-series and nuisance factors (polynomial detrending and tCompCor). In whole-brain voxels, smoothing was applied with a kernel radius of 8 voxels (18.24 μm). Then, a Tukey taper (taper size = 8) was used for GLM pre-whitening [63]. A mixed-effects model [43] was used for group analysis, and the t-value of each voxel was calculated via the second-level analysis of OLS regression with a Tukey taper. These analyses were implemented using in-house MATLAB scripts [64].

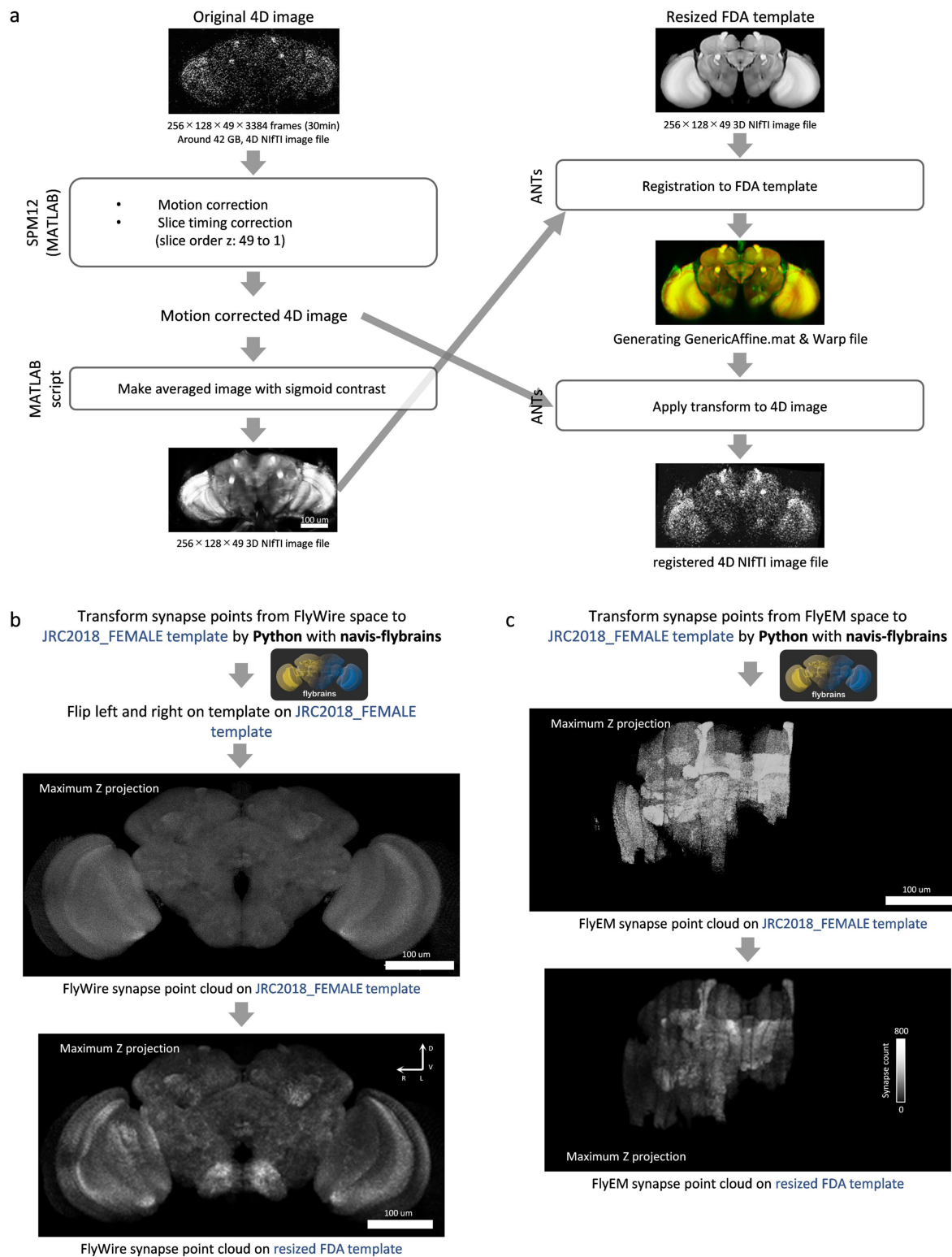
Statistical Analysis

A Mann–Whitney U test was used to examine whether the combination of spatial smoothing and nuisance factor removal was effective. Statistical significance was set at $p<0.05$, and Bonferroni correction was applied to correct the familywise error (FWE) rate. A permutation test was performed to analyze whether specific synapses (segregated, unsegregated, or reciprocal synapses) affect SC-FC correlation. The SC matrix (specific synapses) in the 63 ROIs in the hemibrain was generated, and its SC-FC correlation was calculated. Then, 99 null SC matrices (random synapses) were generated so that they would be normally distributed around the total number of connections in the SC matrix (specific synapses). Since the SC-FC correlations of the null SC matrices (random synapses) also roughly followed a normal distribution, the T-value of the SC-FC correlation of the SC matrix (specific synapses) was calculated using the cumulative distribution function of the normal distribution. For GLM analysis, a mixed-effects model was used for group analysis.

Data availability

Whole-brain calcium imaging data and walking behavior data in *Drosophila Melanogaster* can be downloaded from DANDI web site (<https://dandiarchive.org/dandiset/000727/0.240106.0043>). FlyWire connectome data can be downloaded from flywire codex (<https://codex.flywire.ai/>) and Zenodo web site (<https://zenodo.org/records/10676866>). FlyEM connectome data can be downloaded from Janeria's DVID web site (<https://dvid.io/blog/release-v1.2/>).

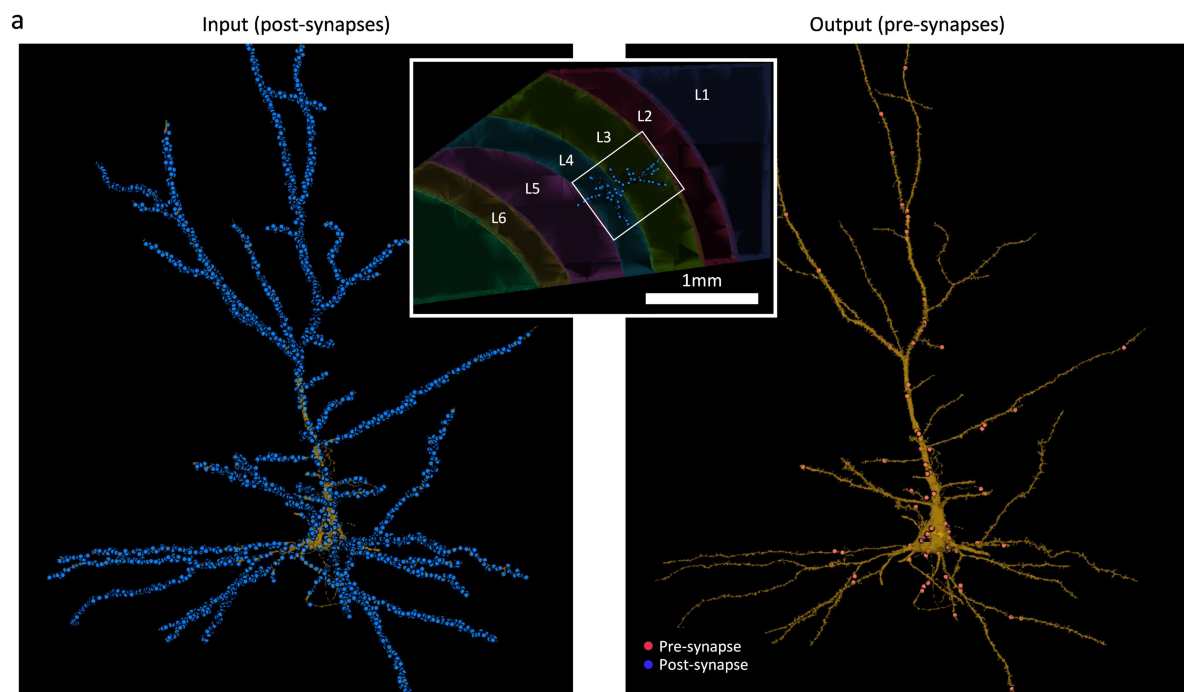
Extended data figures



Extended Data Fig.1-1

Schematic image of registration process and connectome data transformation in *Drosophila melanogaster*.

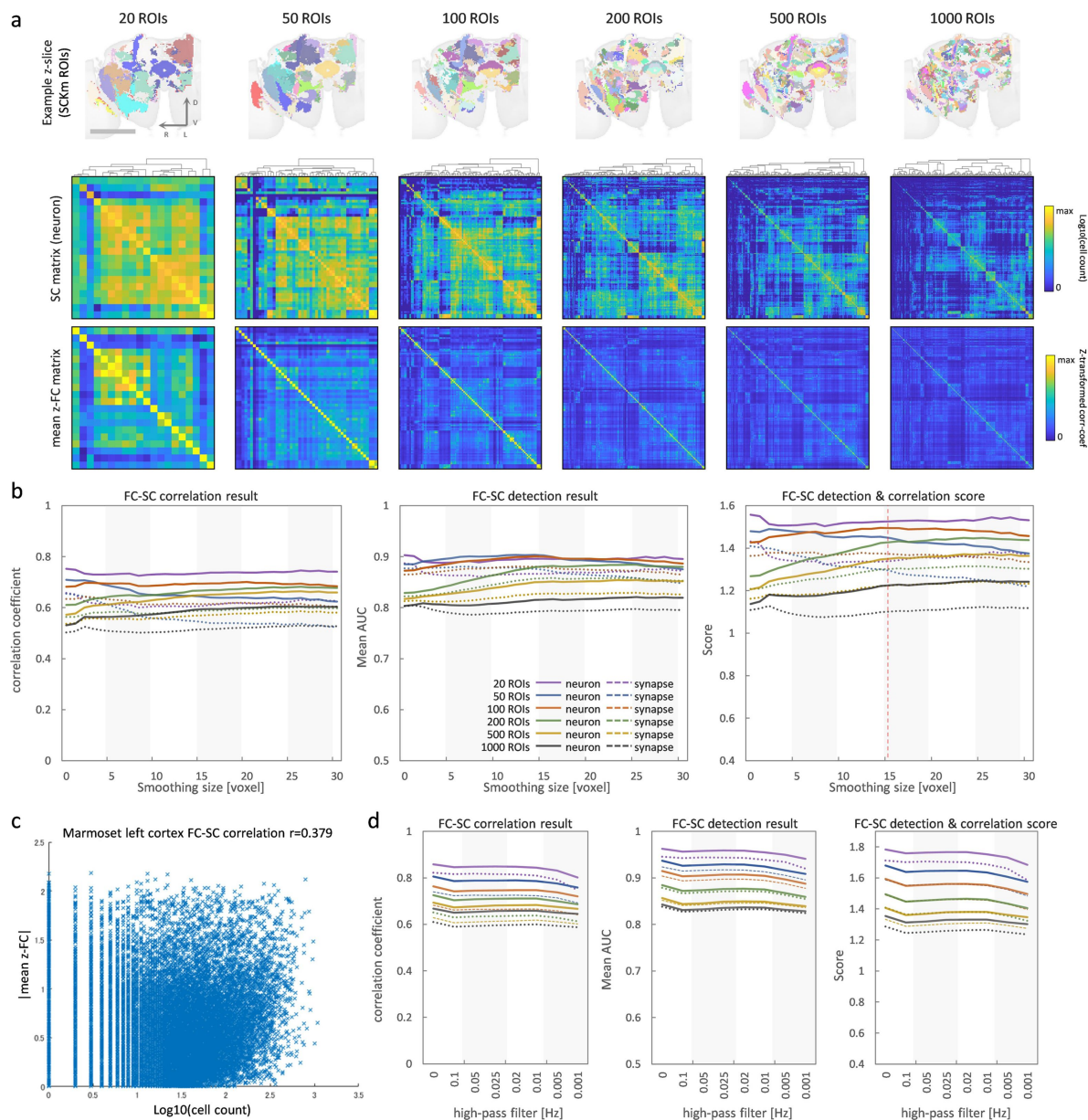
a, Schematic image of calcium image registration to the resized FDA template. Motion and slice timing correction were conducted using SPM12 [57], and the averaged NIfTI image was registered by ANTs [58] to the resized FDA template. Lastly, 4D NIfTI images were transformed using the transformation information. **b**, Schematic image of synapse points



Extended Data Fig.1-2

Pre- and Post-synapses of pyramidal cell in human temporal cortex layer 3-4.

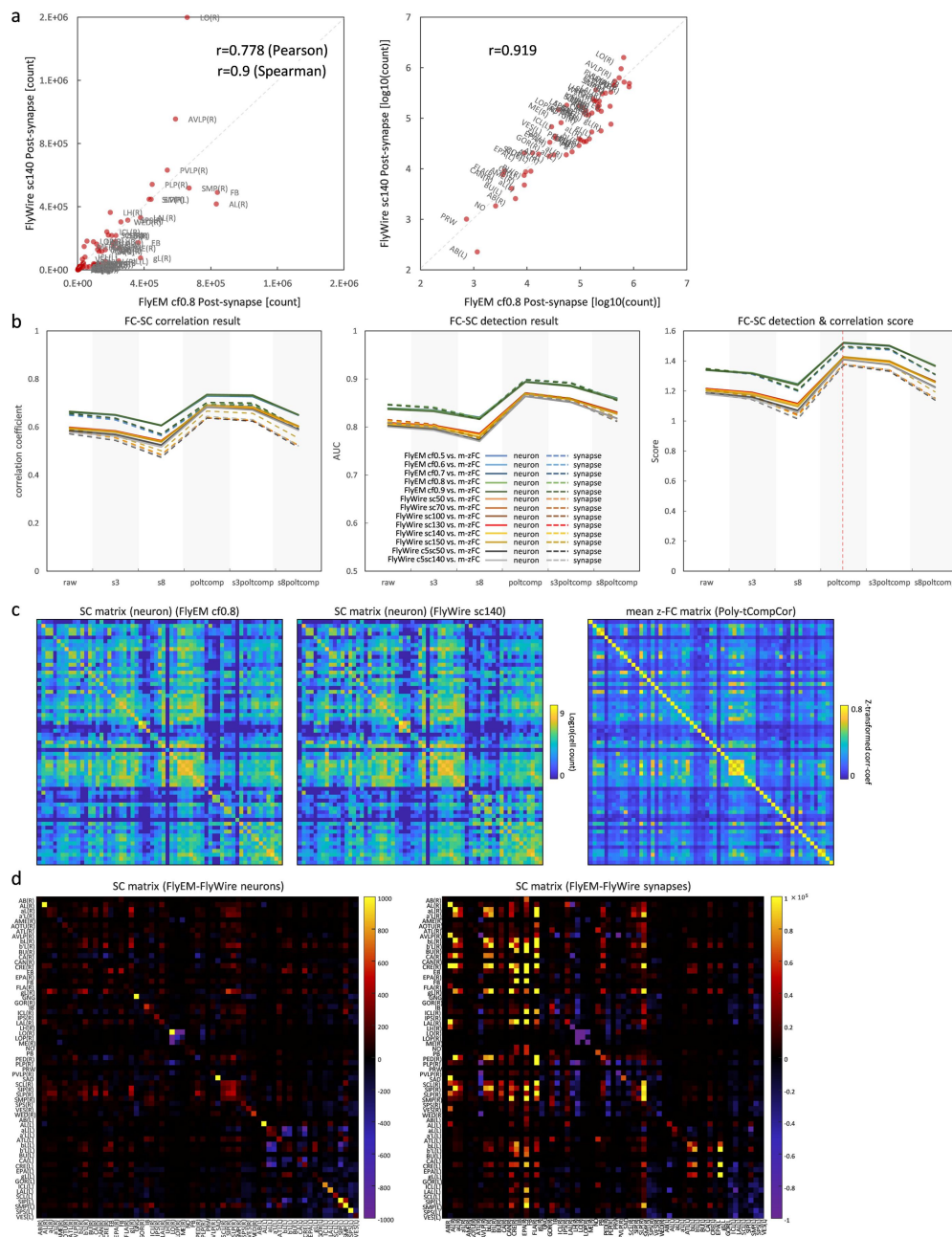
a, (left) Input to (post-synapses) a pyramidal cell in the human temporal cortex, from EM connectome data [39]. (center insert) A slice of human temporal cortex and target pyramidal cell. (right) Output from (pre-synapses) a pyramidal cell in the human temporal cortex, from EM connectome data. Blue dots show post-synaptic sites, Red dots show pre-synaptic sites.



Extended Data Fig.2-1

Investigating spatial smoothing size, and high-pass filtering.

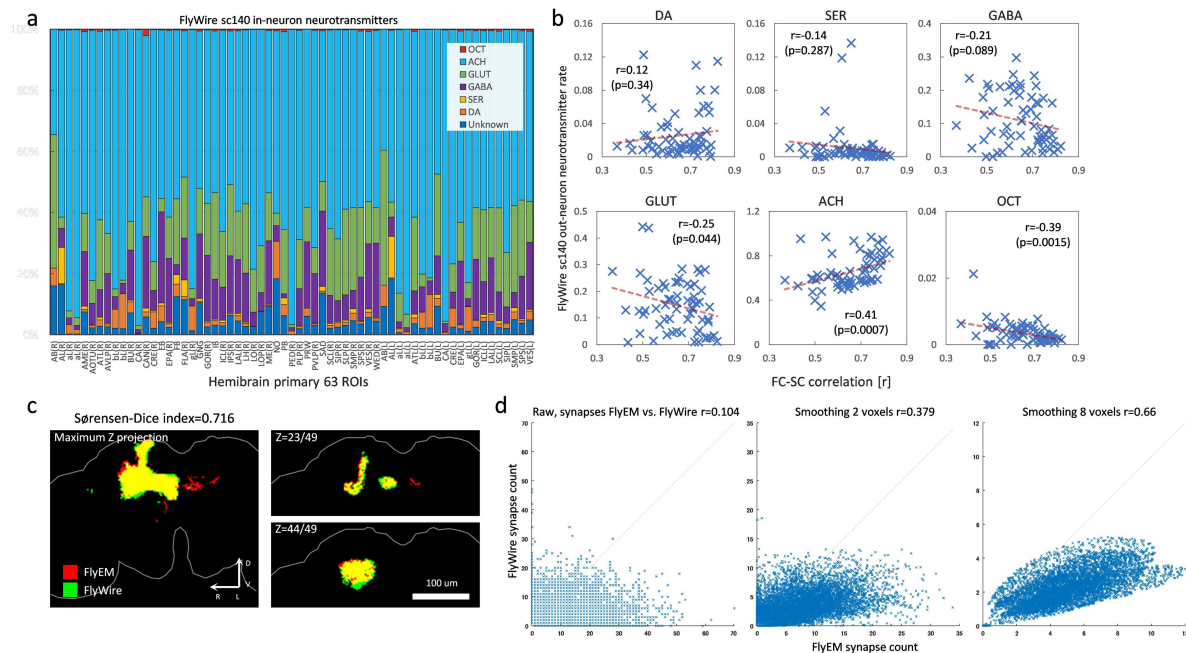
a, (top) Example z-slice of reduced FDA template with SCKm ROIs from 20 to 1000 ROIs. (middle) SC matrices (neuron) of SCKm ROIs from 20 to 1000 ROIs. (bottom) FC matrices of SCKm ROIs from 20 to 1000 ROIs. **b**, From left to right, FC-SC correlation, FC-SC detection, and FC-SC detection & correlation score, respectively. Solid lines shows FC vs. SC matrix (neuron), and dashed lines shows FC vs. SC matrix (synapse) for each ROI count. The vertical axis shows the correlation coefficient, averaged AUC, and FC-SC detection & correlation score, respectively. The horizontal axis shows spatial smoothing size (voxels). **c**, Scatter plot of FC-SC correlation in the marmoset left cortex [10] (145 injection $\times$ 9862 voxels, $r=0.379$). **d**, Result of high-pass filtering and nuisance factor removal combination of DistKm ROIs. From left to right, FC-SC correlation, FC-SC detection, and FC-SC detection & correlation score, respectively. The high-pass filter degraded the FC-SC relationship.



Extended Data Fig.3-1

Comparison between FlyEM and FlyWire connectome data.

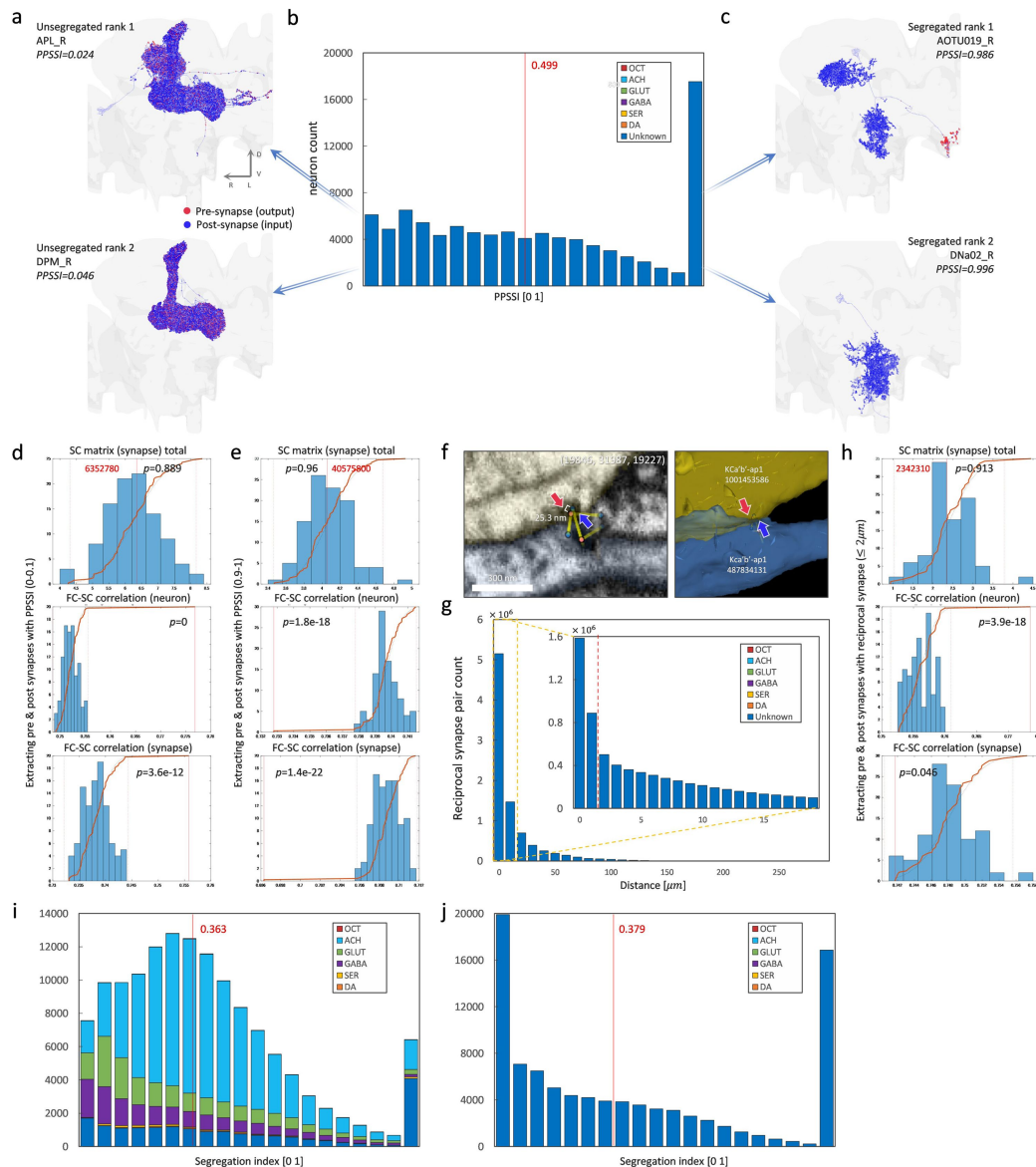
a, (left) A scatter plot of regional post-synapse count between FlyEM (cf0.8) and FlyWire (sc140) in the hemibrain primary 63 ROIs. The Pearson correlation is not so similar, $r=0.778$, but the Spearman correlation is much more similar, $r=0.9$. (right) A scatter plot of log10 of the regional post-synapse count between FlyEM (cf0.8) and FlyWire (sc140) in the hemibrain primary ROIs. Pearson correlation shows $r=0.919$, a higher similarity. **b**, From left to right, FC-SC correlation, FC-SC detection, and FC-SC detection & correlation score, respectively. Solid lines shows FC vs. SC matrix (neuron), and dashed lines shows FC vs. SC matrix (synapse) of FlyEM (cf0.5-0.9) and FlyWire (sc50-140) corresponding Fig.3d. The vertical axis shows the correlation coefficient, averaged AUC, and FC-SC detection & correlation score, respectively. The horizontal axis shows spatial smoothing & nuisance factor removal combinations. **c**, From left to right, SC matrix (neuron) (FlyEM cf0.8), SC matrix (neuron) (FlyWire sc140) and FC matrix (Poly-tCompCor, no smoothing) in the hemibrain primary 63 ROIs. **d**, (right) SC matrix (neuron) result of FlyEM (cf0.8) minus FlyWire (sc140) in hemibrain primary ROIs. (left) SC matrix (synapse) result of FlyEM (cf0.8) minus FlyWire (sc140). FlyEM shows a higher number of synaptic connections.



Extended Data Fig.3-2

In-neuron neurotransmitters and comparison between FlyEM and FlyWire connectome data.

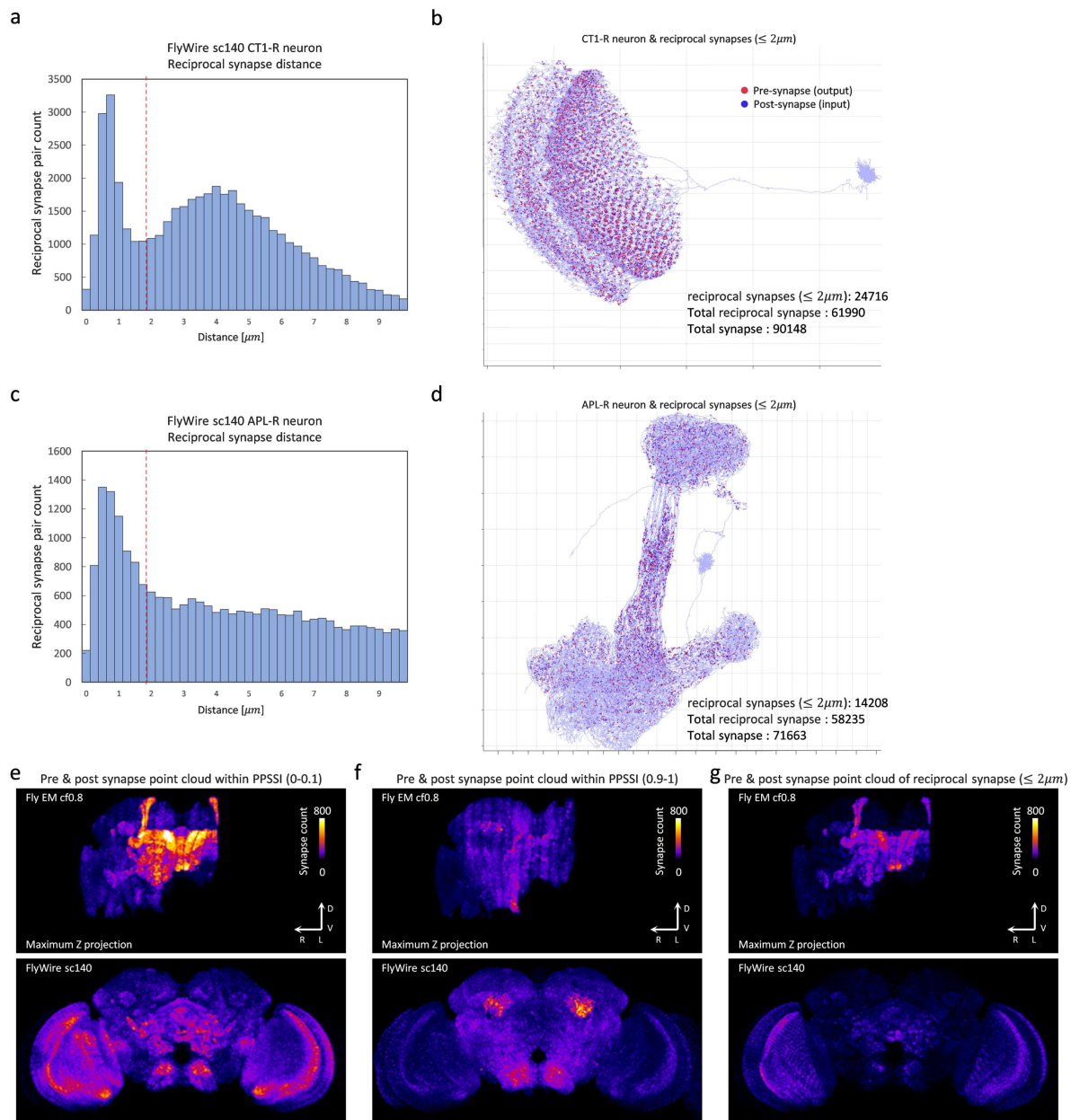
a, Bar graph of neurotransmitter rate of input neurons in hemibrain primary 63 ROIs based on FlyWire (sc140) connectome data [44]. The horizontal axis shows hemibrain primary ROIs. (DA: dopamine, SER: serotonin, GABA, GLUT: glutamine, ACH: acetylcholine, OCT: octopamine) **b**, Scatter plots of neurotransmitter rate of input neurons vs. FC-SC correlation (neuron) in hemibrain primary ROIs (FlyWire sc140). Each of the six neurotransmitters was compared. **c**, Comparison between FlyEM and FlyWire APL right neuron (synapse point cloud mask). Red voxels are FlyEM, green voxels are FlyWire, and yellow voxels are both. Sørensen-Dice index (0.716) shows similarity, so neuropil shape is similar between two. **d**, (left) A scatter plot of post-synapse count between FlyEM and FlyWire APL right neuron. Pearson correlation shows $r=0.104$, so not similar at all. (center) Same as left, but synapse point cloud was spatially smoothed by gaussian kernel (2 voxels). Similarity was increased ($r=0.379$). (right) Same as center, but smoothed by gaussian kernel (8 voxels). Similarity was increased ($r=0.66$). These results support that spatial smoothing can absorb inter-individual variability.



Extended Data Fig. 4-1

Quantification of pre- and post-synapse segregation and relationship between FC and synapses (FlyEM).

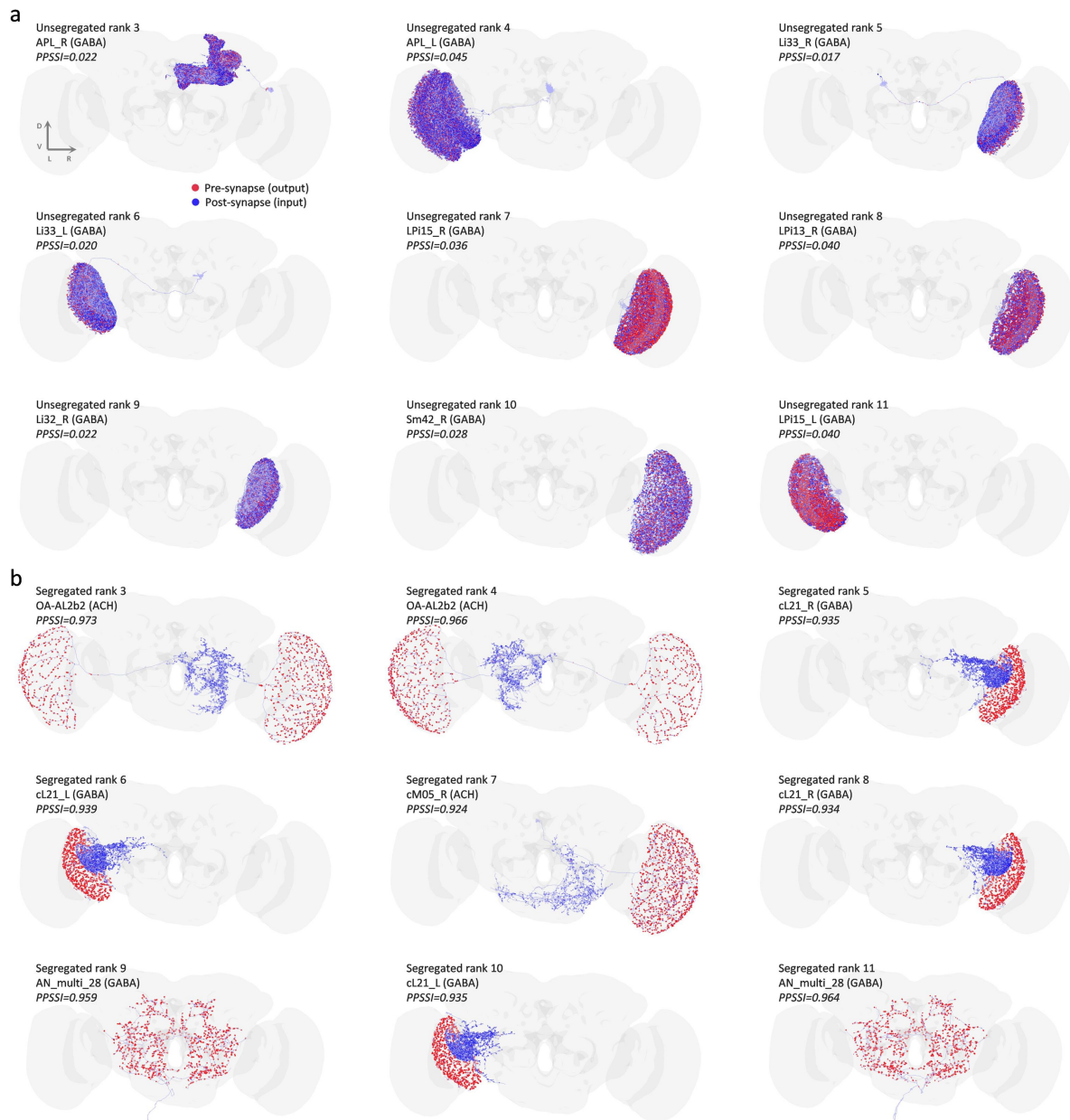
a, Unsegregated first (APL_L cell) and second (DPM_R cell) ranked cells in the FlyEM (cf0.8) connectome data. Red dots shows pre-synapses, blue dot shows post-synapses. **b**, The PPSSI histogram of hemibrain neurons (99,644) in FlyEM (cf0.8) connectome data. The vertical axis shows neuron count, and the horizontal axis shows PPSSI [0 1]. **c**, Segregated first (AOTU019_R cell) and second (DNa02_R cell) ranked cells in the FlyEM (cf0.8) connectome data. **d**, Histogram of null SC matrices (blue bar) and extracted SC matrix with PPSSI (0-0.1) (red solid line). The black dotted line shows the cumulative distribution function of the normal distribution, the brown solid line shows the cumulative distribution function of null & extracted SC matrices, and the red dotted line shows the Bonferroni-corrected $p < 0.05$ threshold. Top shows the SC matrix (synapse) total, middle shows FC-SC correlation (neuron), and bottom shows FC-SC correlation (synapse). **e**, Histogram of null SC matrices and extracted SC matrix with PPSSI (0.9-1). **f**, Example of a reciprocal-synapse in the EM image (left) and 3D image (right). **g**, Histogram of reciprocal-synaptic minimum distances in the FlyEM (cf0.8) connectome data. The vertical axis shows reciprocal-synapse pair count, and the horizontal axis shows distance [μm]. **h**, Histogram of null SC matrices and extracted SC matrix with reciprocal-synapses ($\leq 2\mu\text{m}$). **i**, The segregation index histogram of all neurons (139,255) in FlyWire (sc140) connectome data. The horizontal axis shows segregation index [0 1]. **j**, The segregation index histogram of hemibrain neurons (99,644) in the FlyEM (cf0.8) connectome data. The horizontal axis shows the segregation index [0 1].



Extended Data Fig.4-2

Reciprocal synapses of CT1 and APL neurons, and synapse point cloud for several conditions.

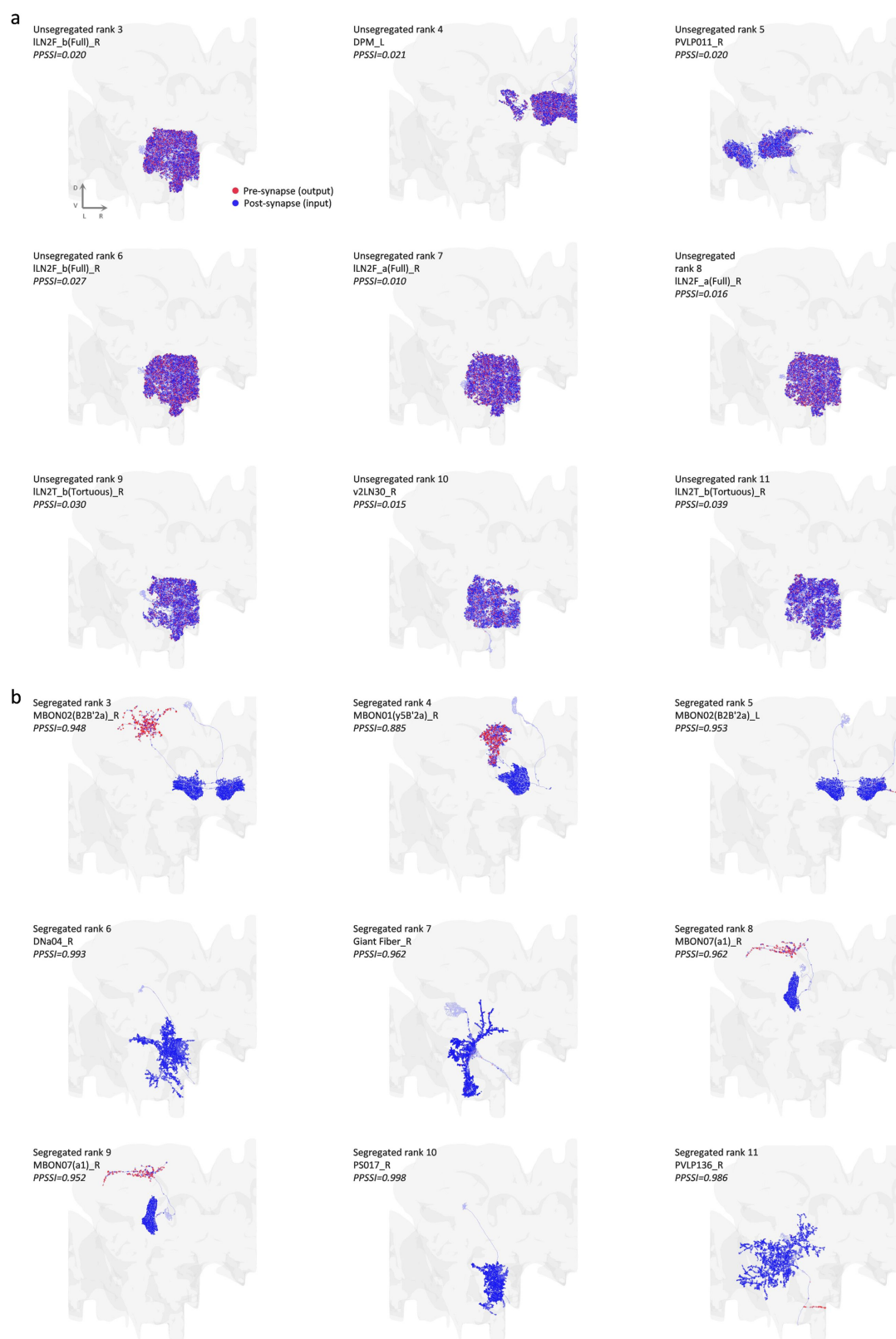
a, Histogram of reciprocal-synaptic minimum distances of the CT1-R neuron in the FlyWire (sc140) connectome data. The vertical axis shows reciprocal-synapse pair count, and the horizontal axis shows distance [μm]. **b**, Skeleton view of the CT1-R neuron. Red dots are pre-synapses and blue dots are post-synapses of reciprocal synapses ($\leq 2\mu\text{m}$). **c**, Histogram of reciprocal-synaptic minimum distances of the APL-R neuron in the FlyWire (sc140) connectome data. **d**, Skeleton view of the APL-R neuron, the same way as **b**. **e**, (top) Maximum Z-projection of pre & post synapse point cloud within PPSSI (0-0.1) in the FlyEM data (cf0.8). (bottom) Maximum Z-projection of pre & post synapse point cloud within PPSSI (0-0.1) in the FlyWire data (sc140). **f**, The same as in **e**, but with the condition PPSSI in 0.9-1. **g**, The same as in **e**, but with the condition of only reciprocal synapses ($\leq 2\mu\text{m}$).



Extended Data Fig.4-3

Unsegregated and segregated cell ranking examples (FlyWire sc140).

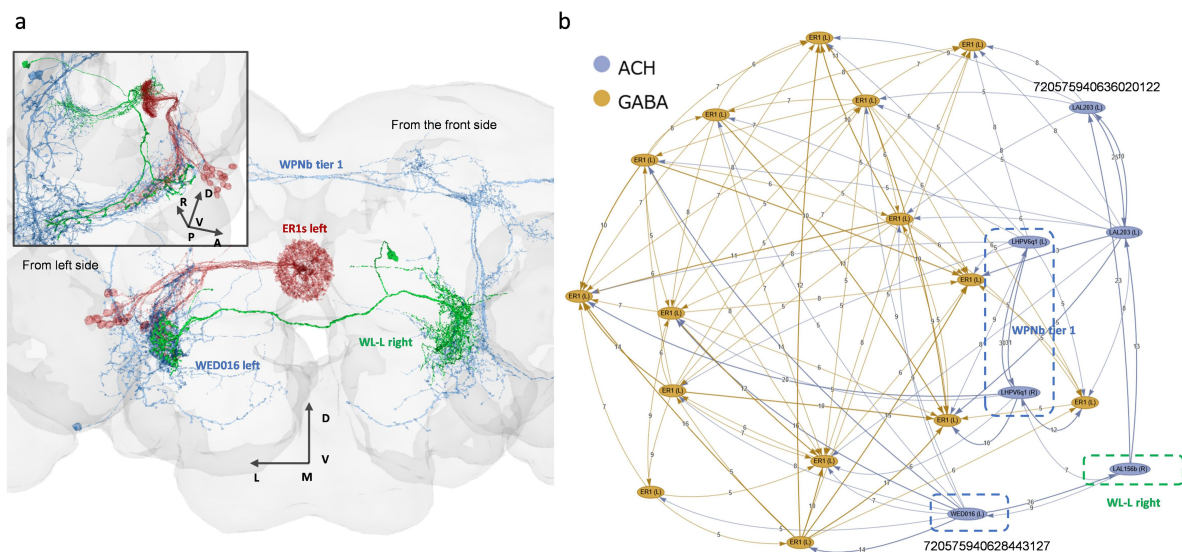
a, Unsegregated cell ranking from 3rd to 11th in the FlyWire_{sc140} connectome data. Red dots shows pre-synapses, blue dots shows post-synapses. **b**, Segregated cell ranking from 3 to 11th.



Extended Data Fig.4-4

Unsegregated and segregated cell ranking (FlyEM cf0.8).

a, Unsegregated cell ranking from 3 to 11th in the FlyE_{rd}M (cf0.8) connectome data. Red dots shows pre-synapse, blue dots show post-synapses. **b**, Segregated cell ranking from 3 to 11th.



Extended Data Fig.5-1

Mechanosensory pathway from Wedge to Ellipsoid Body.

a, 3D image of WL-L right, WED016 left, WPNb tier 1, and ER1 left neurons, frontal view. Black square insert shows WL-L right, WED016 left, WPNb tier 1 and ER1 left neurons from the left lateral view. **b**, Network graph of WL-L right, WED016 left, WPNb tier 1 and ER1 left neurons from the FlyWire codex (sc50, connection threshold $\geq$ 5). Blue is for acetylcholine, other for GABA neurons. The small number on the edges shows synapse count.

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Additional information

Code Availability

Software and code used in the current study is open source and publicly available from <https://github.com/takuto-okuno-riken/flywalk> 

Authors' Contributions

T.O. conceived of the presented idea. T.O. developed the theory, performed the computations. T.O. downloaded and processed fly data. T.O., A.W., H.O. and J.H. discussed the results and contributed to the final manuscript.

Ethics declarations

All experimental procedures were approved by the Experimental Animal Committee of RIKEN or by the Experimental Animal Committee of the National Center of Neurology and Psychiatry. The marmosets were handled by the “Guiding Principles of the Care and Use of Animals in the Field of Physiological Science” formulated by the Japanese Physiological Society.

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Reviewer #1 (Public review):

Summary:

In this paper, the authors analyze connectome data from *Drosophila* and compare the physical wiring with functional connectivity estimated from calcium imaging data. They quantify structure-function relationships as a correlation of the two connectivity modalities. They report correlations roughly comparable to what has been described in the literature on sc/fc relationships in mammalian connectome data at the meso-scale. They then repeat their analysis, focusing on segregated versus unsegregated synapses. They derive separate connectomes using one or the other class of synapse. They show differential contributions to the sc/fc relationships by segregated versus unsegregated synapses.

Strengths:

There is nice synthesis of multimodal imaging data (Ca and EM data from flies and meso-scale data from human and marmoset).

Weaknesses:

- (1) The paper is written in an unusual way. The introduction intermingles results with background, making it hard to figure out what precisely is being tested.
- (2) There are also major methodological gaps. Though the mammalian connectomes are used as a point of reference, no descriptions of their origins or processing are included.
- (3) A major weakness stems from the actual calculation of the sc/fc correlation. In general, SC is sparse. In the case of the EM connectomes, it is *exceptionally* sparse (most neural elements are not connected to one another). The authors calculated sc/fc coupling by correlating the off-diagonal elements of sc (the logarithm of its edge weights) and fc matrices with one another. The logarithmic transformation yields a value of infinity for all zero entries. The authors simply impute these elements with 0. This makes no sense and, depending on whether these zero elements are distributed systematically versus uniformly random, could either inflate or deflate the sc/fc correlations. Care must be taken here.
- (4) Further, in constructing the segregated versus unsegregated connectomes, they use absolute thresholds for collecting synapses. It is unclear, however, whether similar numbers of synapses were included in both matrices. If the number is different, that might explain the differential relationship with fc; one matrix has more non-zero entries (and as noted earlier, those zero entries are problematic).
- (5) There was also considerable text (in the results) describing the processing of the Ca data. In this section, the authors frequently refer to some pipelines as "better" or "worse" (more or less effective). But it is not clear what measures they adopted to assess the effectiveness of a pipeline.

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Reviewer #2 (Public review):

Summary:

Okuno et al. investigate the structure-function relationship in the fruit fly *Drosophila melanogaster*. To do so, they combine published data from two recent synapse-level connectomes ("hemibrain" and "FlyWire") with a dataset comprising functional whole-brain calcium imaging and behavioural data. First, they investigate the applicability of fMRI pre-processing techniques on data from calcium imaging. They then cross-correlate this pre-processed functional data with structural data extracted from the connectomes, including a comparison to humans. The authors proceed to compare the two connectomes and find significant differences, which they attribute to differences in the accuracy of the synapse detections. Next, they present a novel algorithm to quantify whether neurons are segregated (pre- and postsynapses are spatially separate) or unsegregated (pre- and postsynapses are mixed). Using this approach, they find that unsegregated neurons may contribute more to function than segregated neurons. Applying a general linear model to the functional dataset suggests that activity in two brain areas (Wedge and AVL) is suppressed during walking. The authors identify a GABAergic neuron in the connectome that could be responsible for this effect and suggest it may provide feedback to the fly's "compass" in the central complex.

Strengths:

The study tackles a relevant question in connectomics by exploring the relationship between structural and functional connectivity in the *Drosophila* brain. The authors apply a range of established and adapted analytical methods, including fMRI-style preprocessing and a novel synaptic segregation index. The effort to integrate multiple datasets and to compare across species reflects a broad and methodical approach.

Weaknesses:

The manuscript would benefit from a clearer overarching narrative to unify the various analyses, which currently appear somewhat disjointed. While the technical methods are extensive, the writing is often convoluted and lacks crucial details, making it difficult to follow the logic and interpret key findings. Additionally, the conclusions are relatively incremental and lack a compelling conceptual advance, limiting the overall impact of the work.

(1) The introduction currently contains a number of findings and conclusions that would be better placed in the results and discussion to clearly delineate past findings from new results and speculations.

(2) The narrative would benefit greatly from some clear statements along the lines of "we wanted to find out X, therefore we did Y".

(3) More concise terminology would be helpful. For example, the connectomes are currently referred to as either "hemibrain", "FlyEM", "whole-brain", or "FlyWire".

(4) The abstract claims "a new, more robust method to quantify the degree of pre- and post-synaptic segregation". However, the study fails to provide evidence that this method is indeed more robust than existing methods.

(5) The authors define unsegregated neurons as having mixed pre- and postsynapses in the same space. However, this ignores the neurons' topology: a neuron can exhibit a clearly defined dendrite with (mostly) postsynapses and a clearly defined axon with (mostly) presynapses, which then occupy the same space. This is different from genuinely unsegregated neurons with no distinct dendritic and axonal compartments, such as CT1.

(6) It is not entirely clear where the marmoset dataset originates from. Was it generated for this study? If not, why is there a note in the Ethics Declaration?

(7) On the differences between hemibrain and FlyWire: What is the "18.8 million post-synapses" for FlyWire referring to? The (thresholded) FlyWire synapse table has 130M connections (=postsynapses). Subsetting that synapse cloud to the hemibrain volume still gives ~47M synapses. Further subsetting to only connections between proofread neurons inside the hemibrain volume gives 19.4M - perhaps the authors did something like that? Similarly, the hemibrain synapse table contains 64M postsynapses. Do the 21M "FlyEM" post-synapses refer to proofread neurons only? If the authors indeed used only (post-)synapses from proofread neurons, they need to make that explicit in results and methods, and account for differences in reconstruction status when making any comparisons. For example, the mushroom body in the hemibrain got a lot more attention than in FlyWire, which would explain the differences reported here. For that reason, connection weights are often expressed as, e.g., a fraction of the target's inputs instead of the total number of synapses when comparing connectivity across connectomic datasets. Furthermore, in Figure 3b, it looks like the FlyWire synapse cloud was not trimmed to the exact hemibrain boundaries: for example, the trimmed FlyWire synapse cloud seems to extend further into the optic lobes than the hemibrain volume does.

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Reviewer #3 (Public review):

Summary:

In this manuscript, Okuno et al. re-analyze whole-brain imaging data collected in another paper (Brezovec et al., 2024) in the context of the two currently available *Drosophila* connectome datasets: the partial "FlyEM" (hemibrain) dataset (Scheffer et al., 2020) and the whole-brain "FlyWire" dataset (Dorkenwald et al., 2024). They apply existing fMRI signal processing algorithms to the fly imaging data and compute function-structure correlations across a variety of post-processing parameters (noise reduction methods, ROI size), demonstrating an inverse relationship between ROI size and FC-SC correlation. The authors go on to look at structural connectivity amongst more polarized or less polarized neurons, and suggest that stronger FC-SC correlations are driven by more polarized neurons.

Strengths:

- (1) The result that larger mesoscale ROIs have a higher correlation with structural data is interesting. This has been previously discussed in *Drosophila* in Turner et al., 2021, but here it is quantified more extensively.
- (2) The quantification of neuron polarization (PPSSI) as applied to these structural data is a promising approach for quantifying differences in spatial synapse distribution.

Weaknesses:

One should not score noise/nuisance removal methods solely by their impact on FC-SC correlation values, because we do not know a priori that direct structural connections correspond with strong functional correlations. In fact, work in *C. elegans*, where we have access to both a connectome and neuron-resolution functional data, suggests that this relationship is weak (Yemini et al., 2021; Randi et al., 2023). Similarly, I don't think it's appropriate to tune the confidence scores on the EM datasets using FC-SC correlations as an output metric.

Any discussion of FC-SC comparisons should include an analysis of excitatory/inhibitory neurotransmitters, which are available in the fly connectome dataset. However, here the authors do not perform any analyses with neurotransmitter information.

Comparisons between fly and human MRI data are also premature here. Firstly, the fly connectomes, which are derived from neuron-scale EM reconstructions, are a qualitatively different kind of data from human connectomes, which are derived from DSI imaging of large-scale tracts. Likewise, calcium data and fMRI data are very different functional data acquisition methods-the fact that similar processing steps can be used on time-series data does not make them surprisingly similar, and does not in my view, constitute evidence of "similar design concepts."

The comparison of FlyEM/FlyWire connectomes concludes that differences are more likely a result of data processing than of inter-individual variability. If this is the case, the title should not claim that the manuscript covers individual variability.

The analysis of the wedge-AVLP neuron strikes me as highly speculative, given that the alignment precision between the connectome and the functional data is around 5 microns (Brezovec* et al, PNAS 2024).

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